INT 453: Digital Multimedia

2024-2025

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Lecture 2

Digital Multimedia Hardware

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books

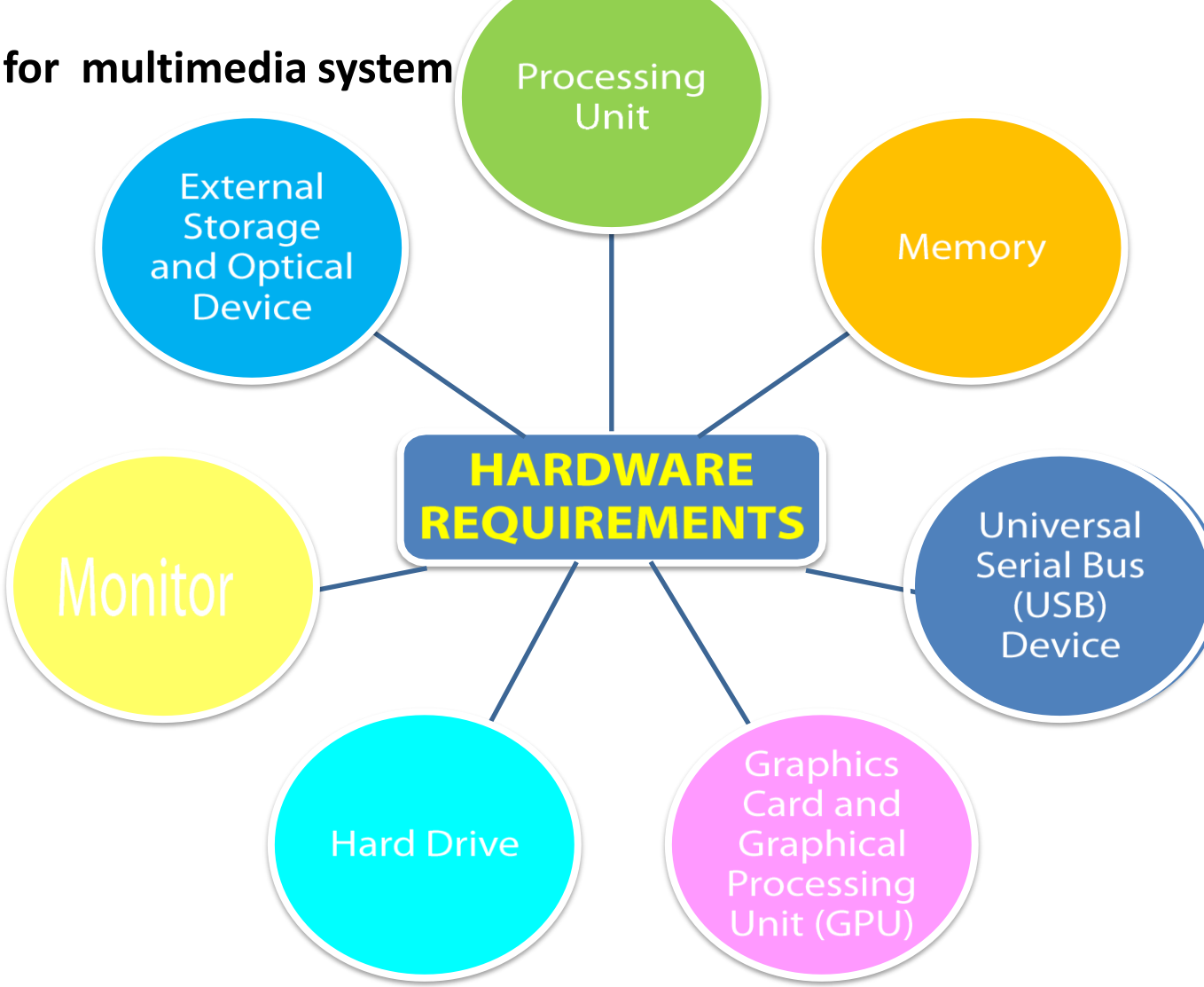
- Multimedia Technologies & applications for the 21st century.
- Fundamentals of Multimedia.

Agenda

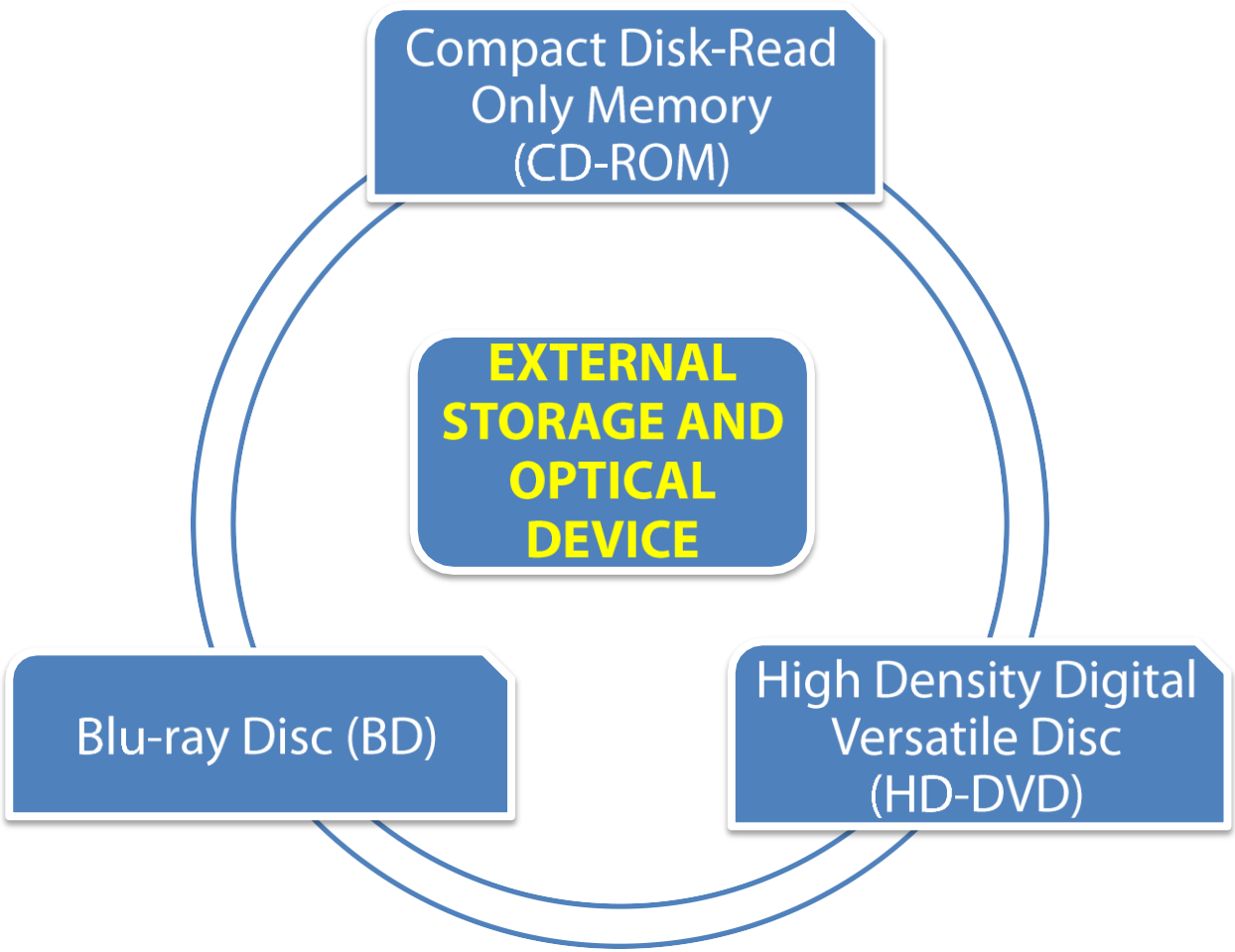
- Discuss the hardware& software requirement of multimedia system
- Introduction to processors
- Understand the Processor architecture for Multimedia
- Multimedia processors classifications with details.
- New Architecture trends for Multimedia processors .

Multimedia Hardware

Hardware requirement for multimedia system

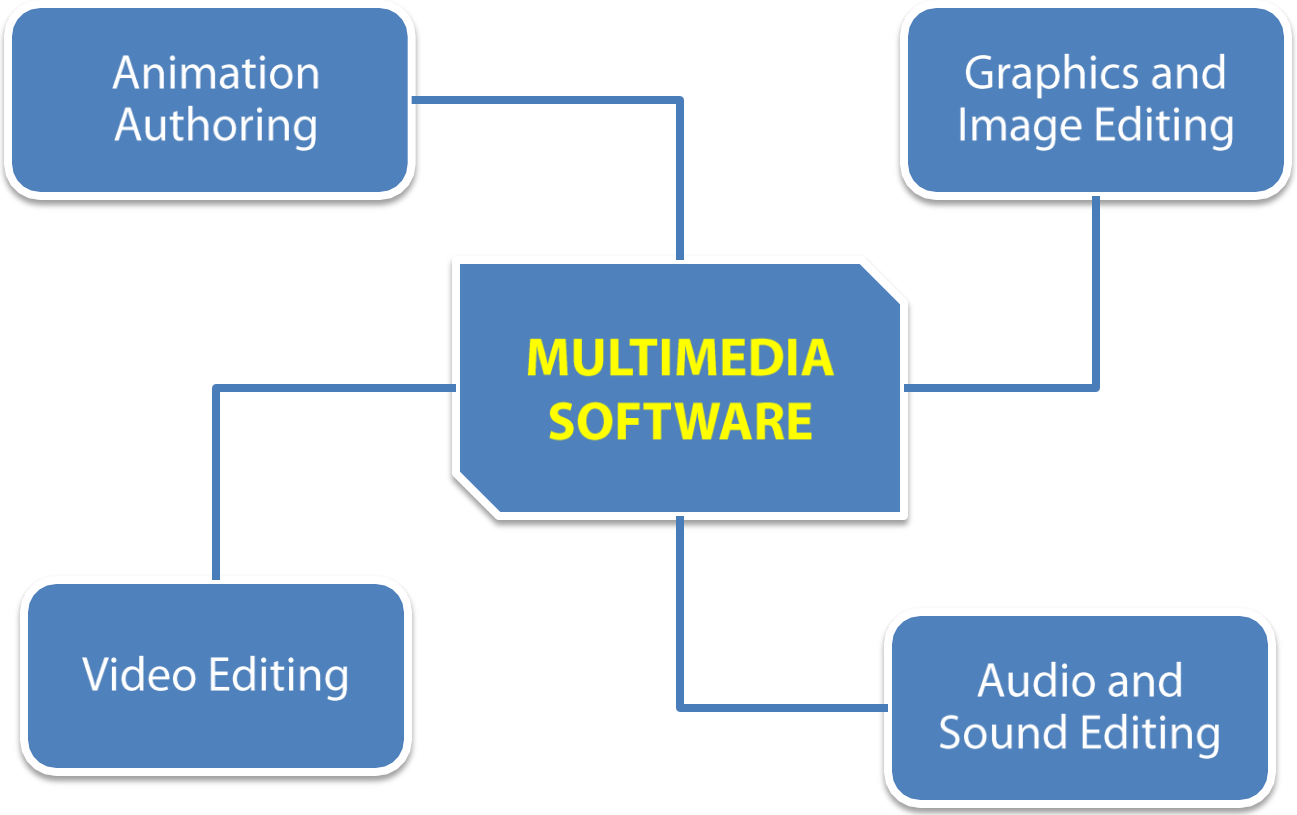


- For displaying quality output images on the monitor we need Graphics Card and Graphical Processing Unit (GPU).



- For saving and distributing multimedia data we need Compact Disk-Read Only Memory (CD-ROM), High Density Digital Versatile Disc (HD-DVD) or Blu-ray Disc (BD).

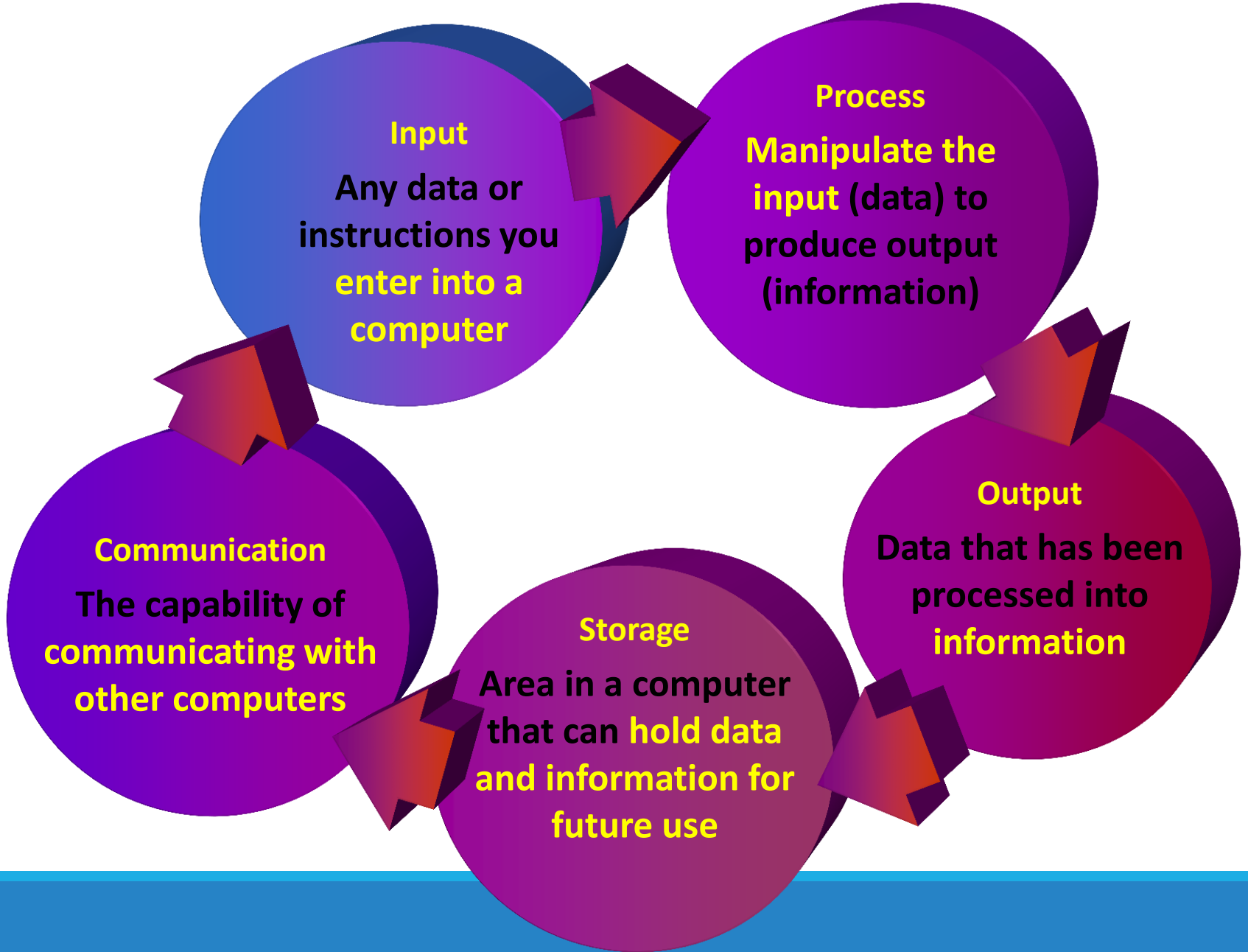
Multimedia Software



- All multimedia data must be in digital format.

Introduction to Processors

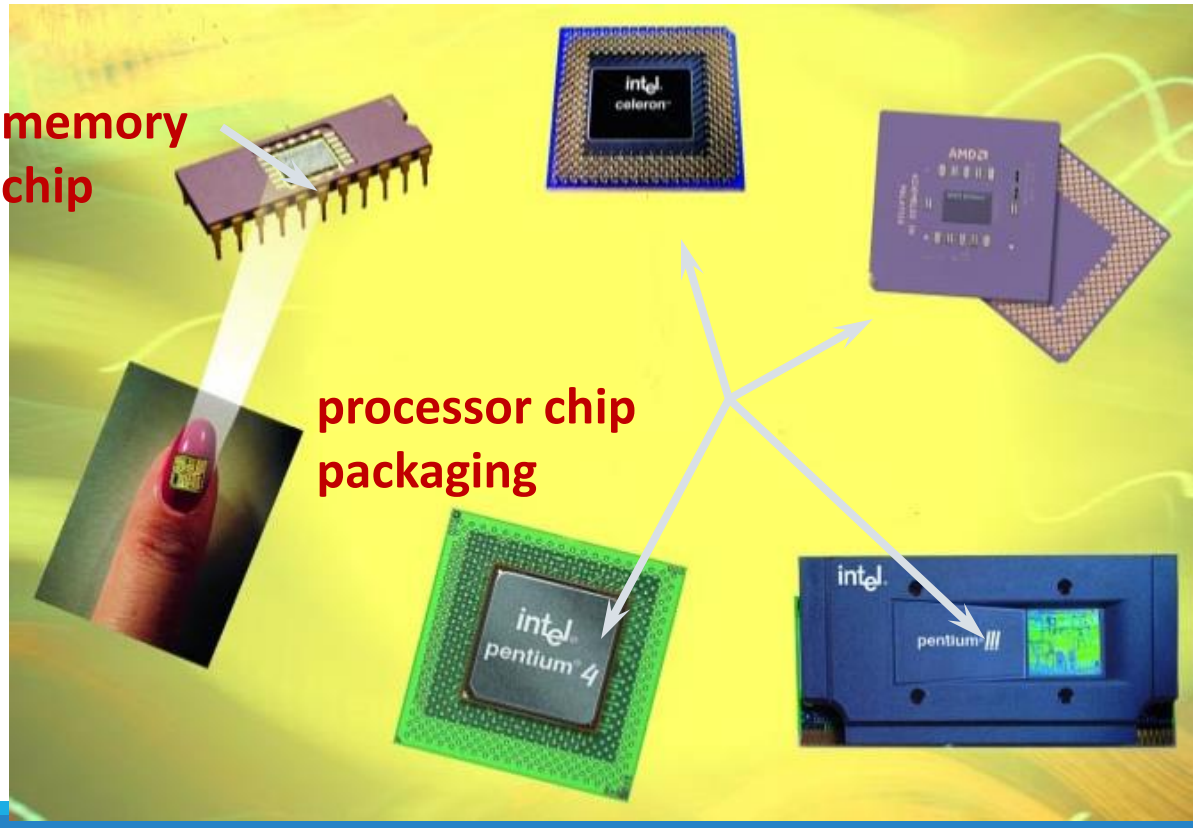
What is the information processing cycle?



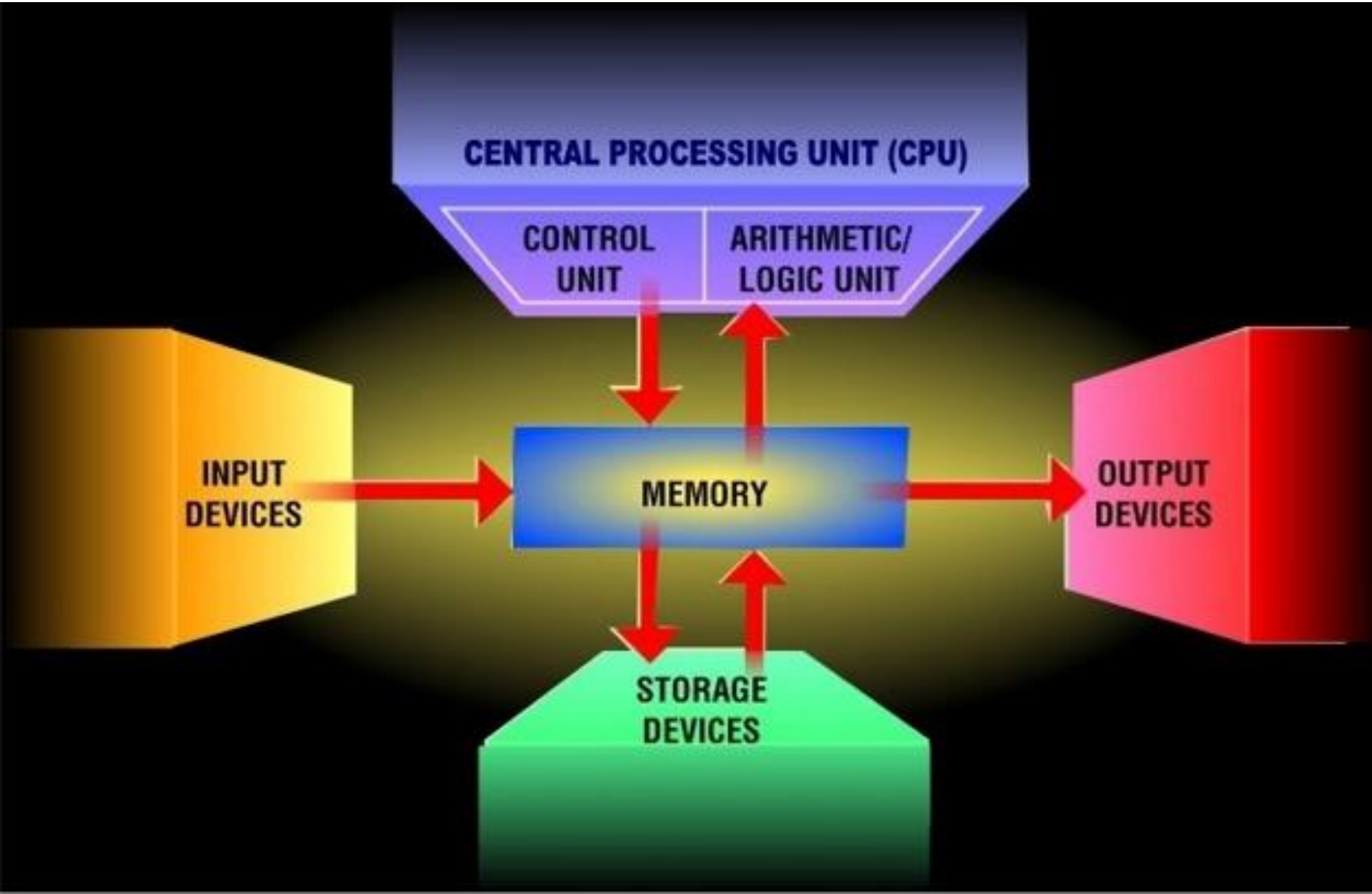
What is a chip?

An electronic device that contains many microscopic pathways that carry electrical currents

- Usually no bigger than one-half inch square
- Packaged so it can be attached to a motherboard



Central Processing Unit



Central Processing Unit

- Also **called the processor**
- Interprets and carries out the basic instructions that operate a computer.
- Most devices communicate with the CPU in order to carry out a task

Inside Microprocessor

Processor consists of the following main parts ;

- Arithmetic and Logic Unit (ALU)

  FPU = floating point unit*: 3D games, Engineering drawings

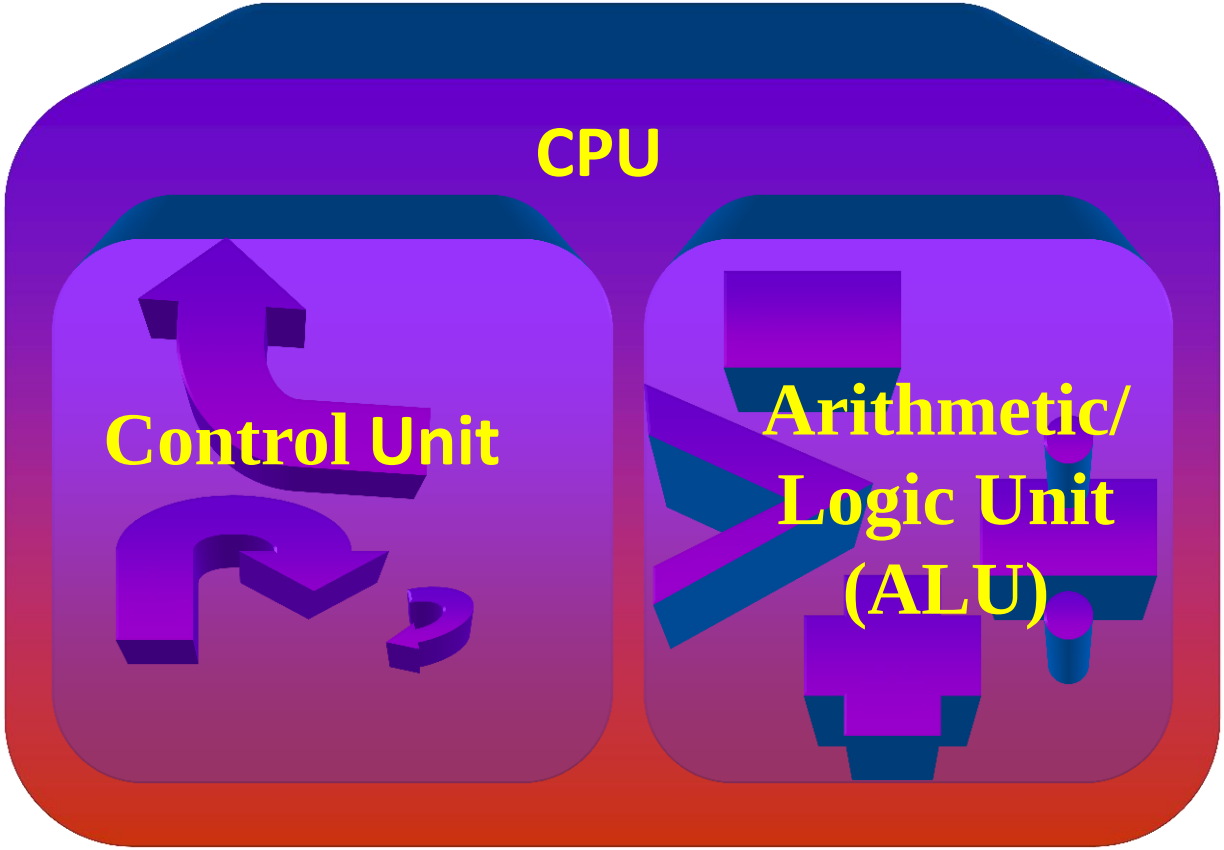
  Registers : SRAM for numbers to be calculated

  Integers unit*: Image processing, 2D games

- Bus interface unit.
- Control unit.
- Cache memory

Central Processing Unit

What are the components of the central processing unit (CPU)?



Central Processing Unit

What is the control unit?



- A component of the CPU that directs and coordinates most of the operations in the computer

The control unit repeats a set of four basic operations:

- Fetch – obtain a program instruction or data item from memory
- Decode - translate the instruction into commands
- Execute - carry out the command
- Store - write the result to memory

Central Processing Unit

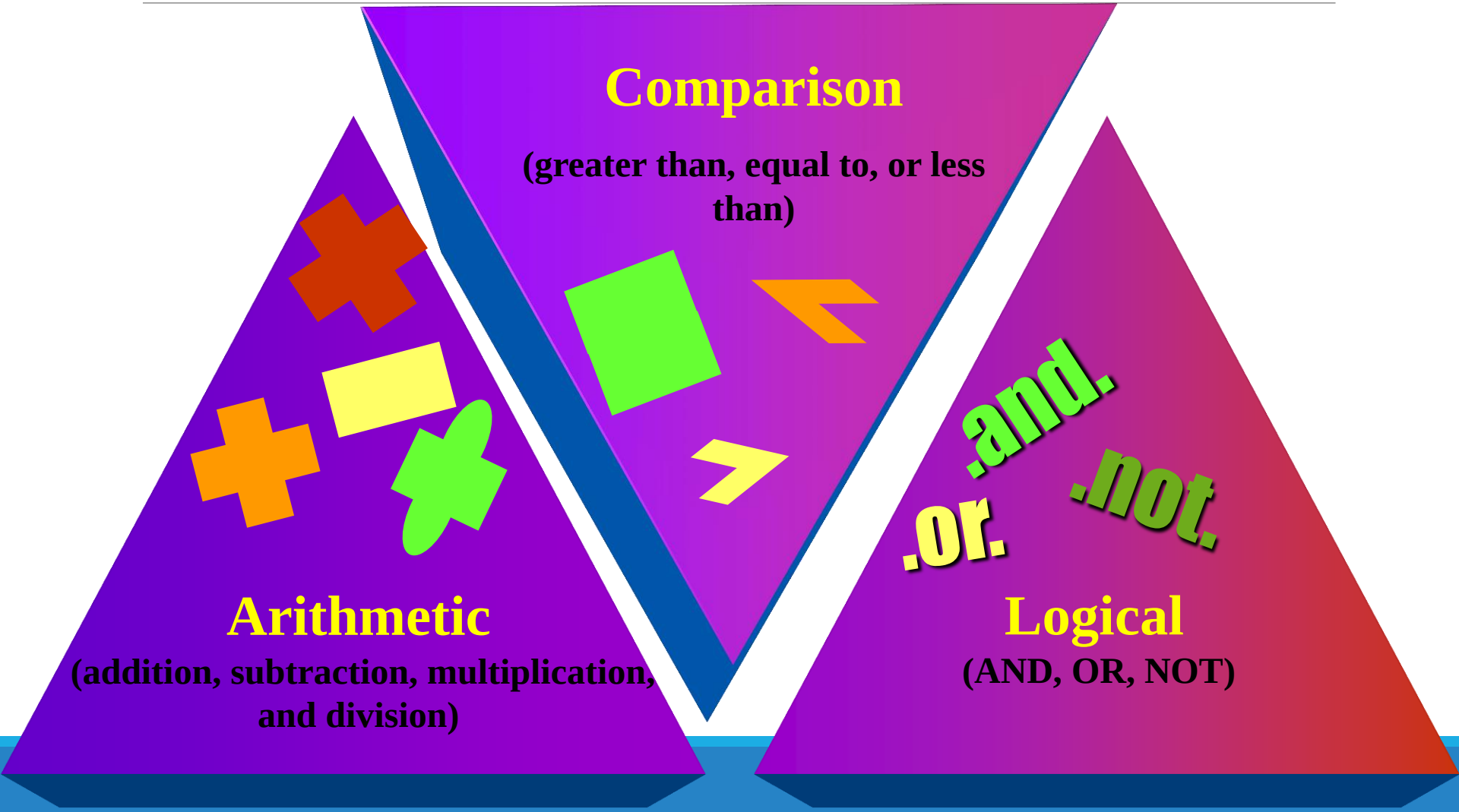
What is the arithmetic/logic unit (ALU)?

- Component of the CPU
- Performs arithmetic, comparison, and logical operations
- Performs the execution part of the machine cycle



Central Processing Unit

What is the arithmetic/logic unit (ALU)?



Central Processing Unit

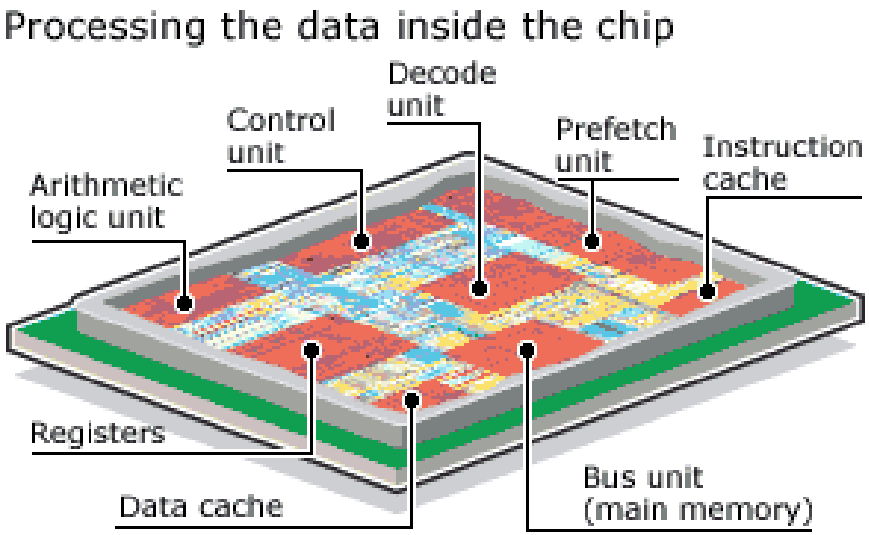
What is a machine cycle?

- Together the four operations of the CPU comprise a machine cycle
- Also called an instruction cycle

- Instruction time (i-time) - time it takes to fetch and decode
- Execution time (e-time) – time it takes to execute and store

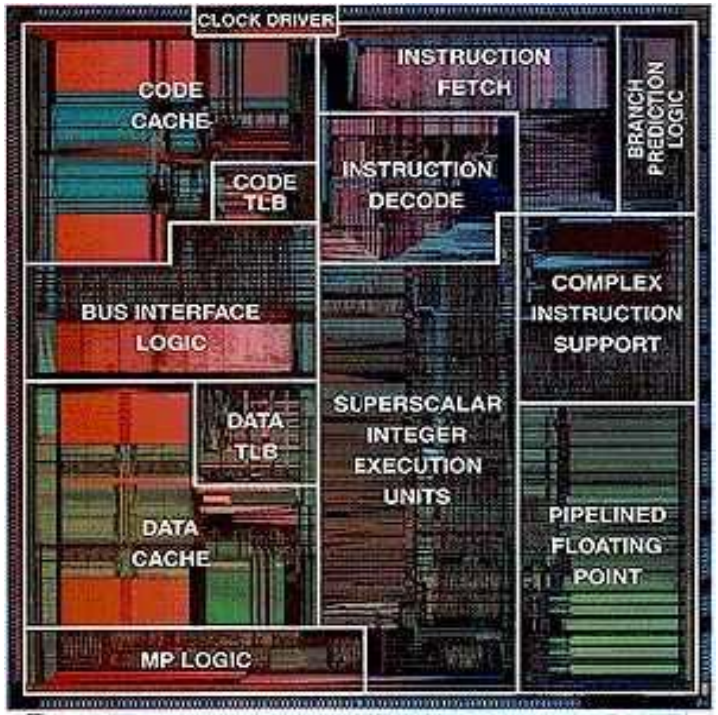


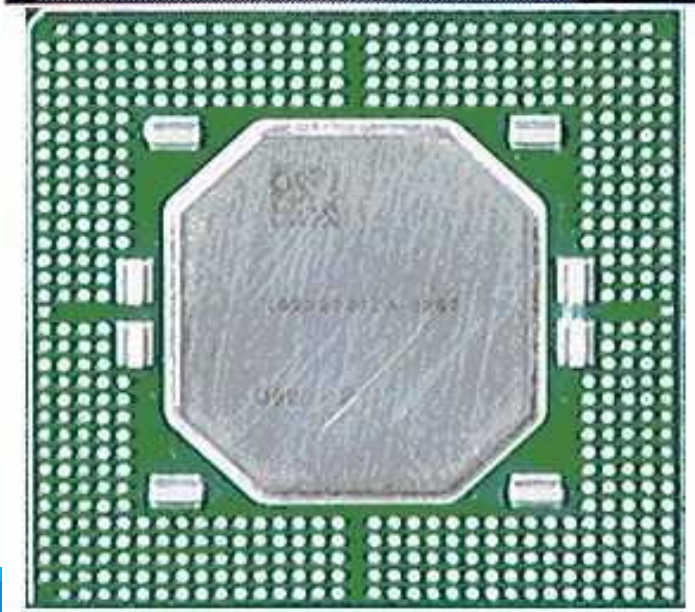
Inside Microprocessor

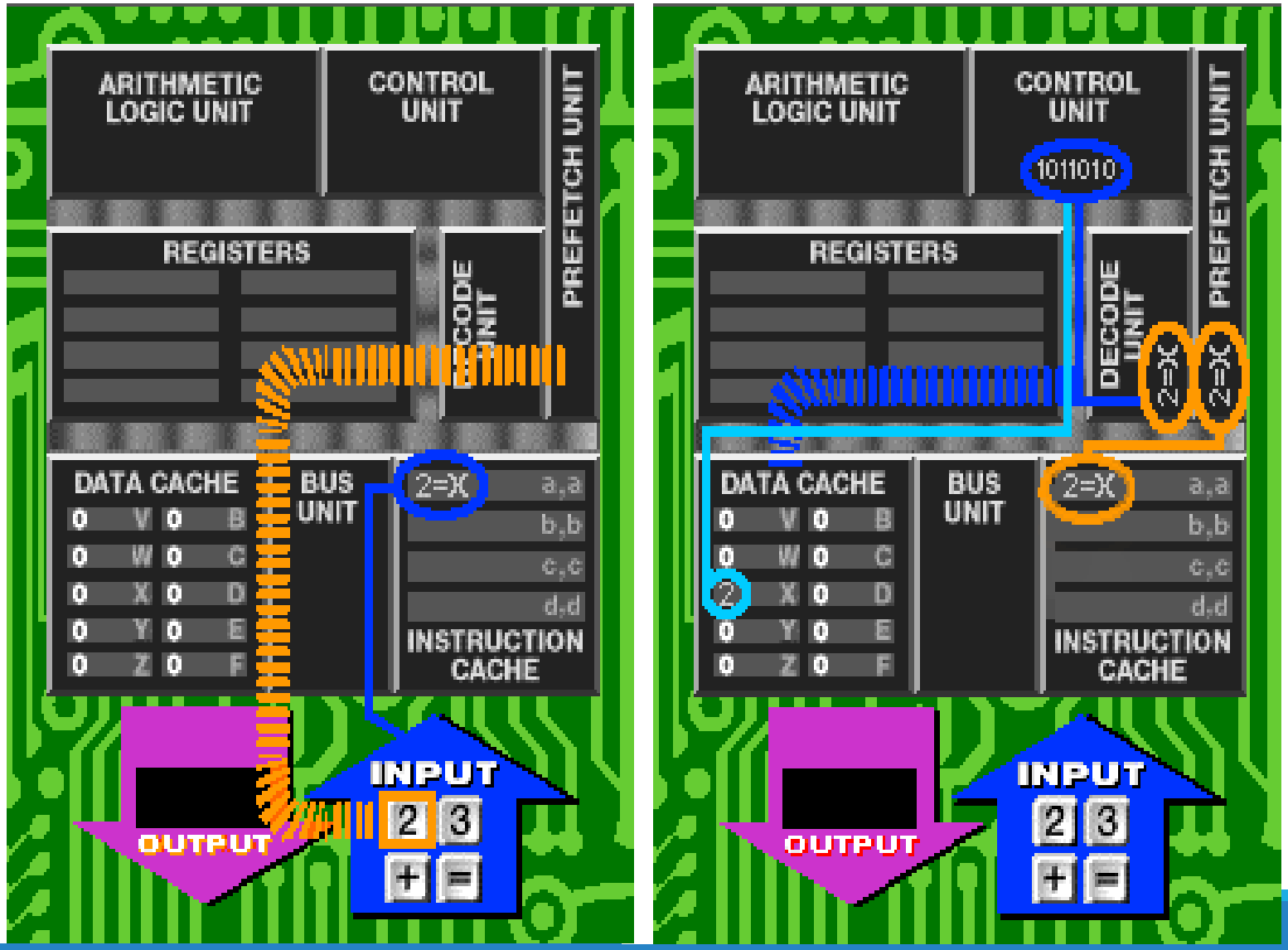




The key processing elements (above) contained in such a tiny chip (right).






ARITHMETIC LOGIC UNIT
CONTROL UNIT
PREFETCH UNIT
REGISTERS
DECODE UNIT
DATA CACHE
BUS UNIT
INSTRUCTION CACHE
INPUT
OUTPUT
1011010

Central Processing Unit

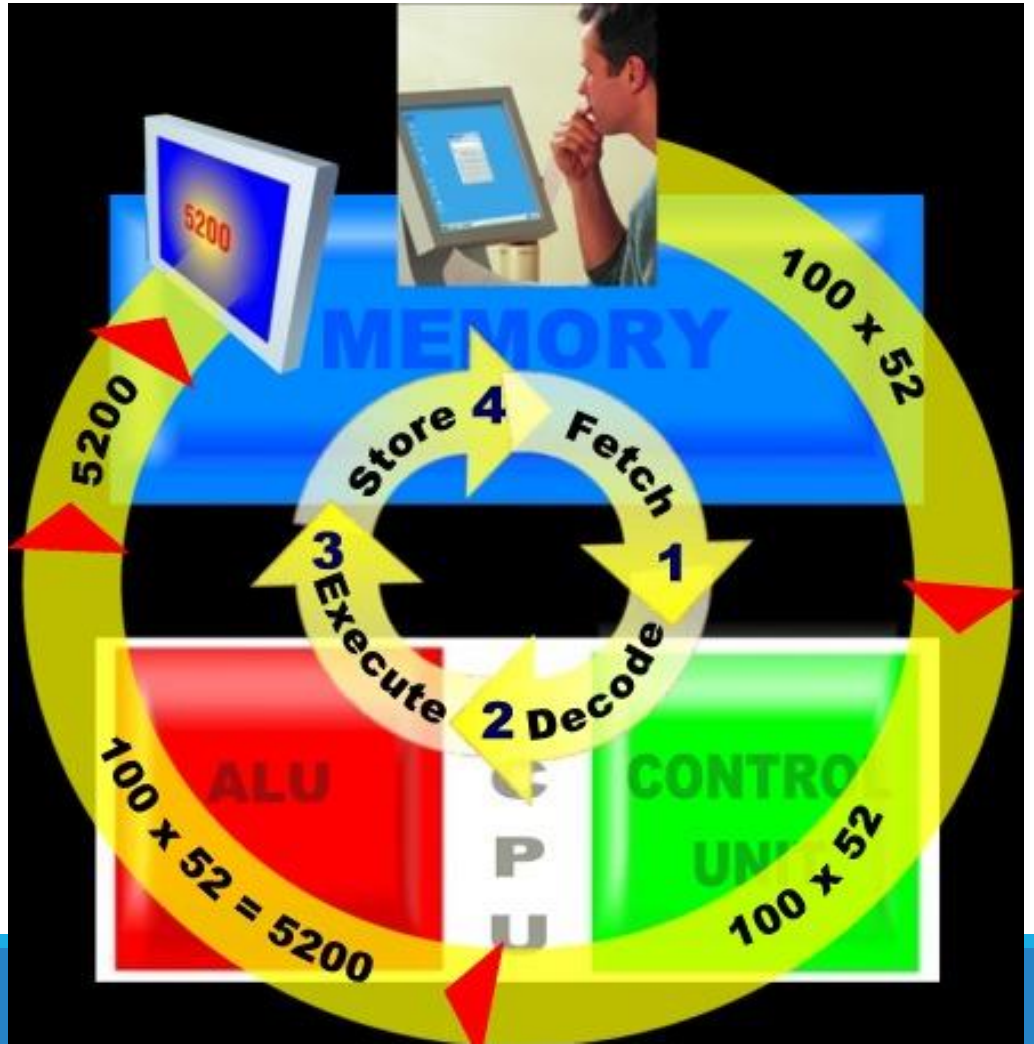
What is a machine cycle?

- **Together the four operations of the CPU comprise a machine cycle**

A student enters a math problem into the memory of the computer.

Step 1: The control unit fetches the math problem from memory.

Step 2: The control unit decodes the math problem and sends it to the ALU.



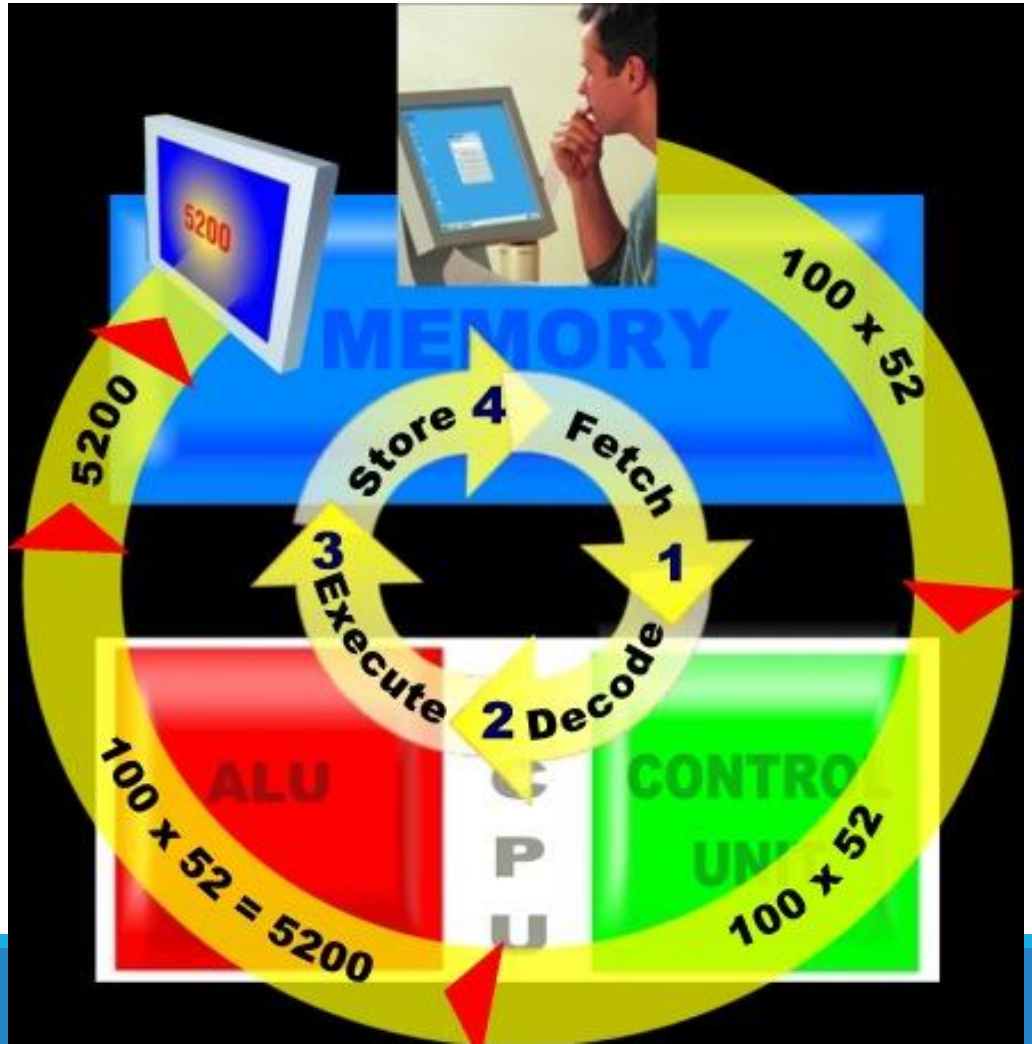
Central Processing Unit

What is a machine cycle?

- **Together the four operations of the CPU comprise a machine cycle**

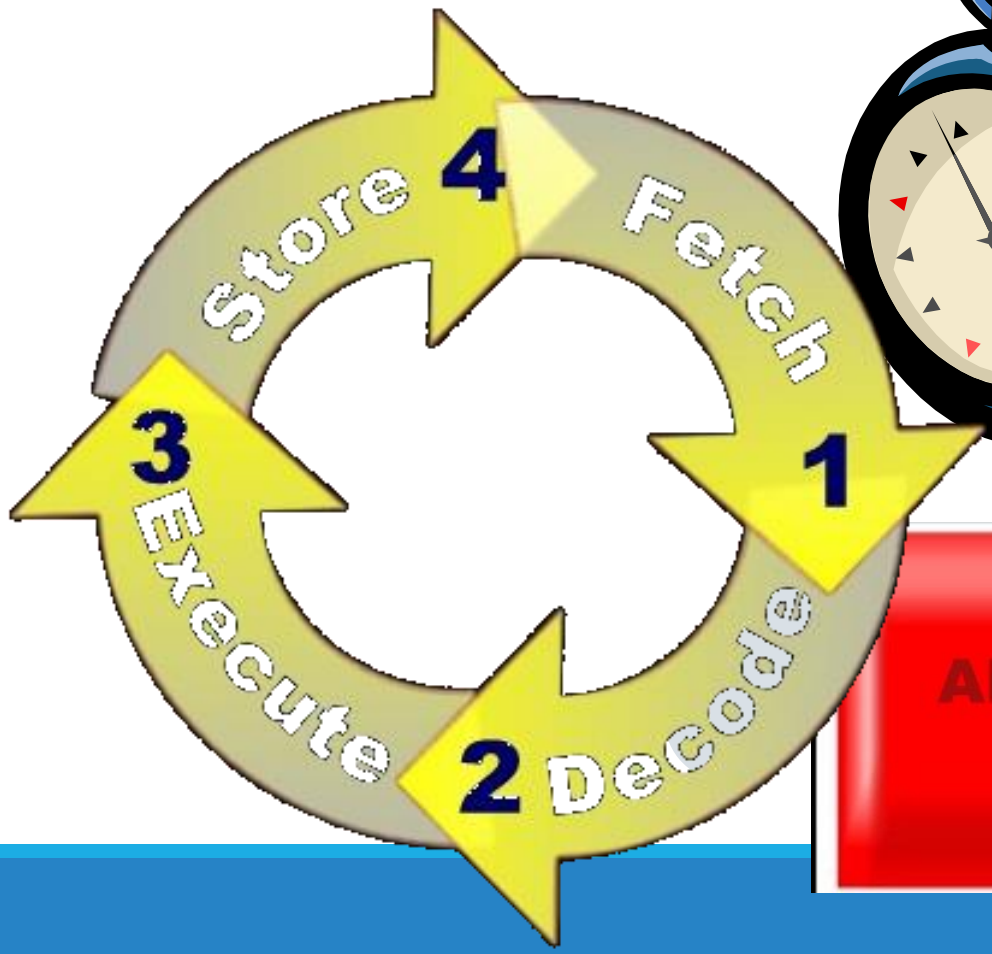
Step 3: The ALU executes the math problem

Step 4: The results of the math problem are stored in memory. The result in memory displays on the screen of the monitor



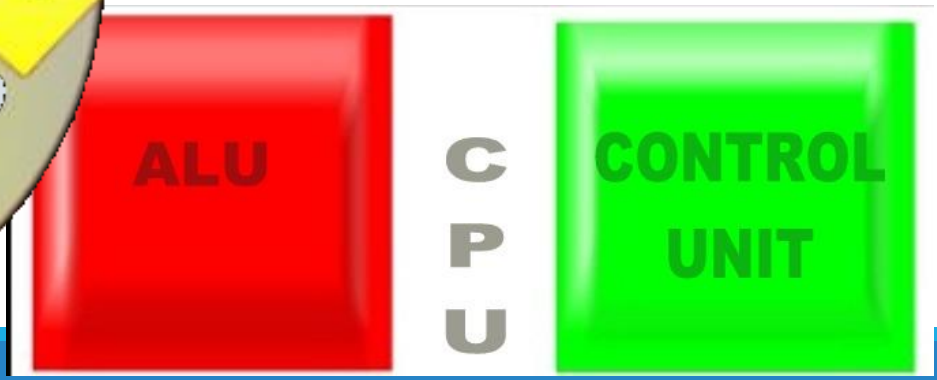
Central Processing Unit

How is the speed of the CPU measured?





- According to how many millions of instructions per second (GIPS) it can process

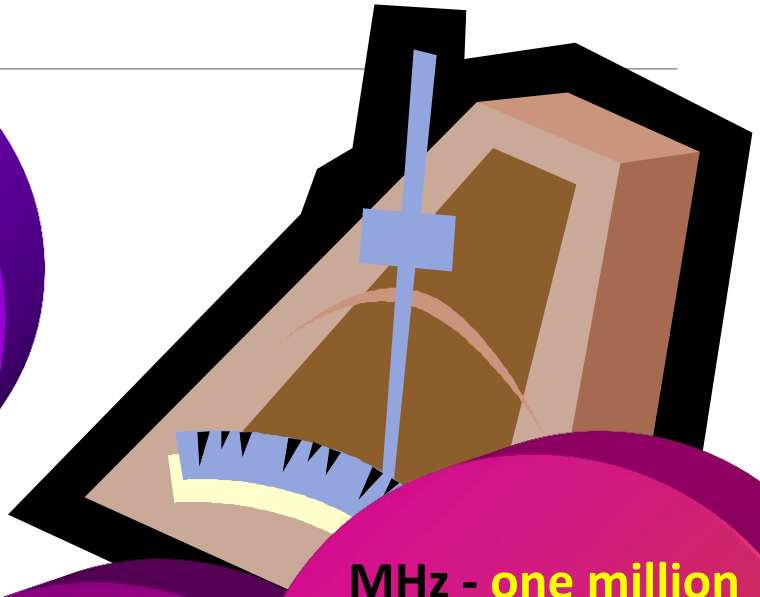


Central Processing Unit

What is the system clock?

Synchronizes all computer operations

Each tick is a clock cycle



Faster clock speed means the CPU can execute more instructions each second

Clock speed (clock rate) measured in megahertz (MHz) and gigahertz (GHz)

MHz - one million ticks of the system clock

GHz – one billion ticks of the system clock

Central Processing Unit

What are two designs used for the CPU?

CISC

(complex instruction set computing)

- **Supports a large number of instructions**
- **CPU executes complex instructions more quickly**

RISC

(reduced instruction set computing)

- **Reduces instructions to only those used more frequently**
- **CPU executes simple instructions more quickly**

Central Processing Unit

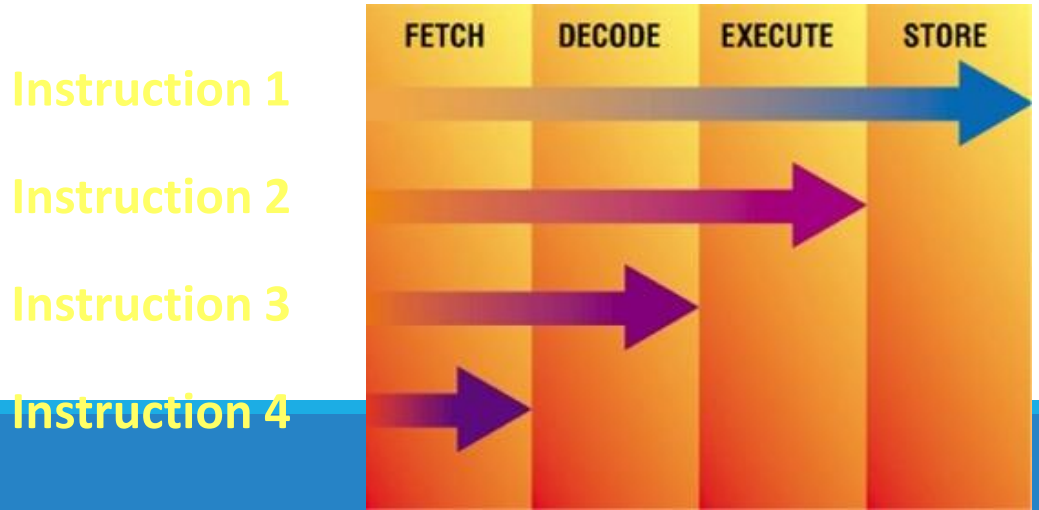
What is pipelining?

- CPU **begins executing a second instruction before completing the first instruction**
- **Results in faster processing**

Machine Cycle (without pipelining)



Machine Cycle (with pipelining)



Central Processing Unit

What is a register?

- Temporary storage location used by the CPU



Cache Memory

- Special group of very fast memory chips located on or close to the CPU
- Level 1 is fastest, then Level 2, then Level 3
- More cache memory typically means faster processing
- Usually internal cache (built into the CPU)

VLSI processor

Very Large-scale Integration (VLSI) is the process of creating an integrated circuit by combining millions or billions of transistors (MOS) into a single chip.

Micro chip creation
https://www.youtube.com/watch?v=_VMYPLXnd7E

CISC vs. RISC

https://www.youtube.com/watch?v=RaFdwNgimKg

Processor Architectures for Multimedia

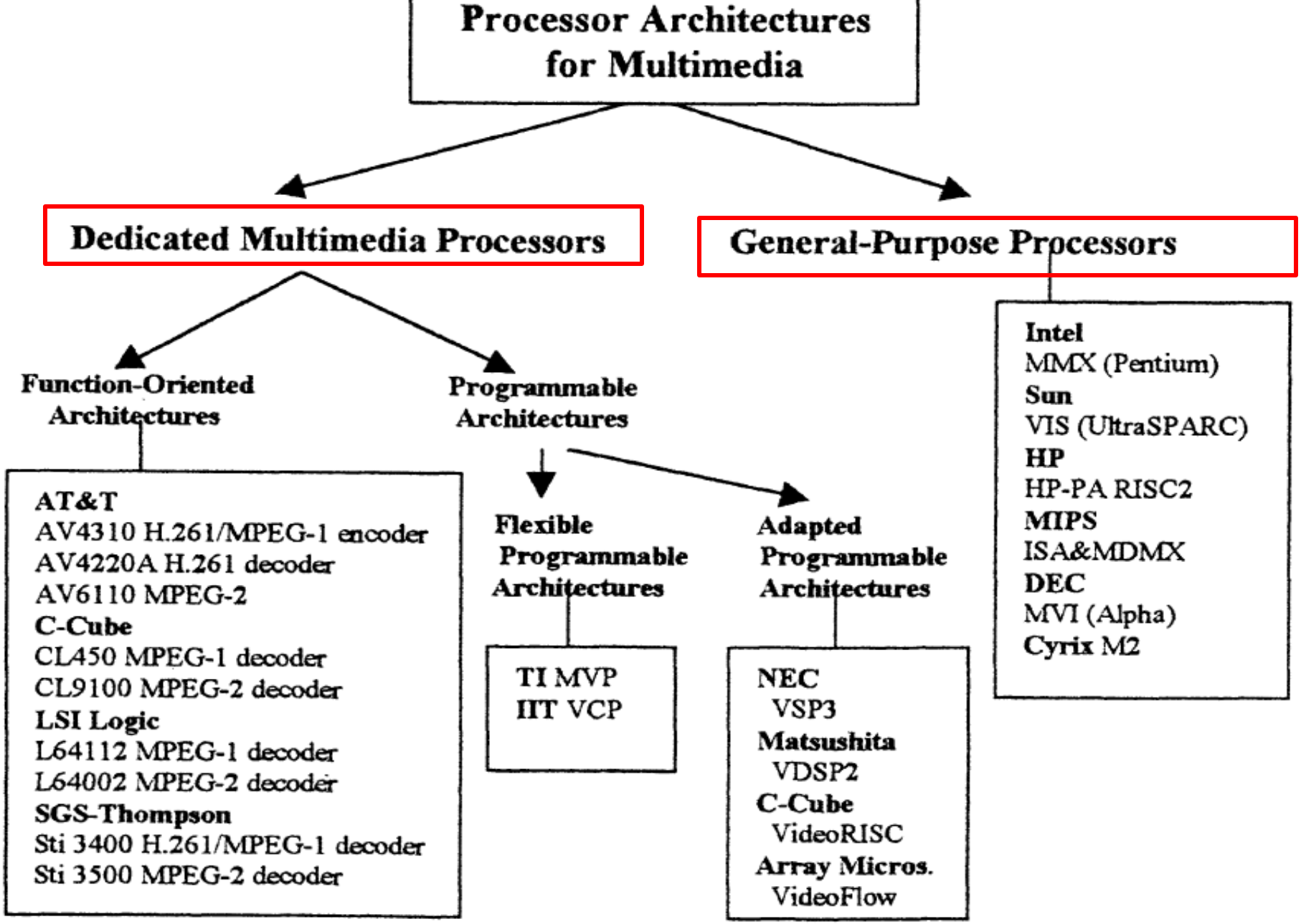


Figure 1. Classification of processor architectures that support multimedia.

Dedicated Multimedia Processors

Dedicated multimedia processors

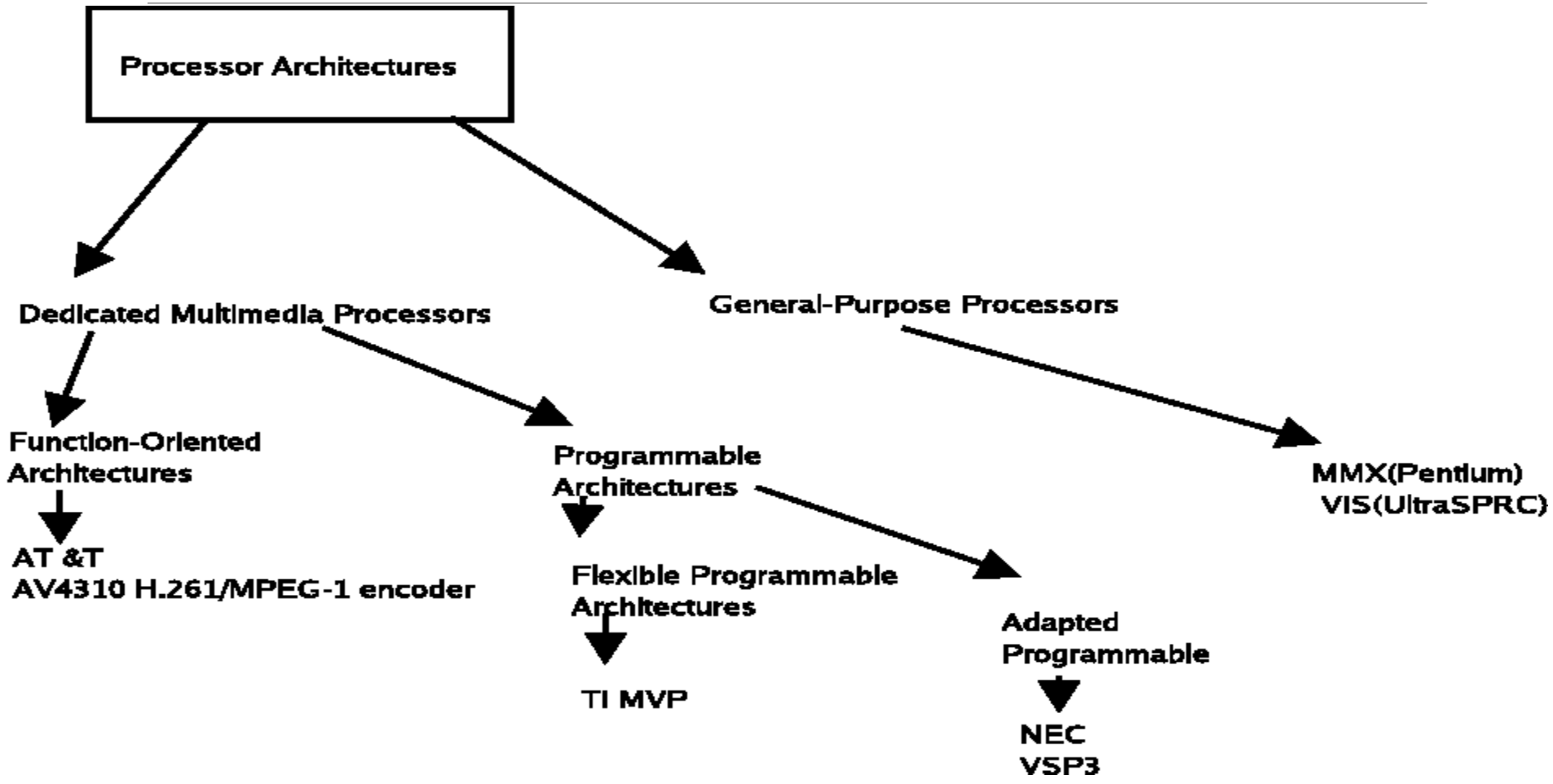
- They are typically custom designed architectures.

- They are intended to perform specific multimedia functions.

These functions usually include:

- Video and audio compression and decompression, and in this case these processors are referred to as video codecs.
- Some advanced multimedia processors provide also support for 2D and 3D graphics applications.
- The general design theme for dedicated multimedia processors consists of using:
  - DSP or RISC core processor for main control, and
  - Special hardware accelerators for the DCT, quantization, entropy encoding, and motion estimation.

Dedicated multimedia processors



Processor Architectures for Multimedia cont.,

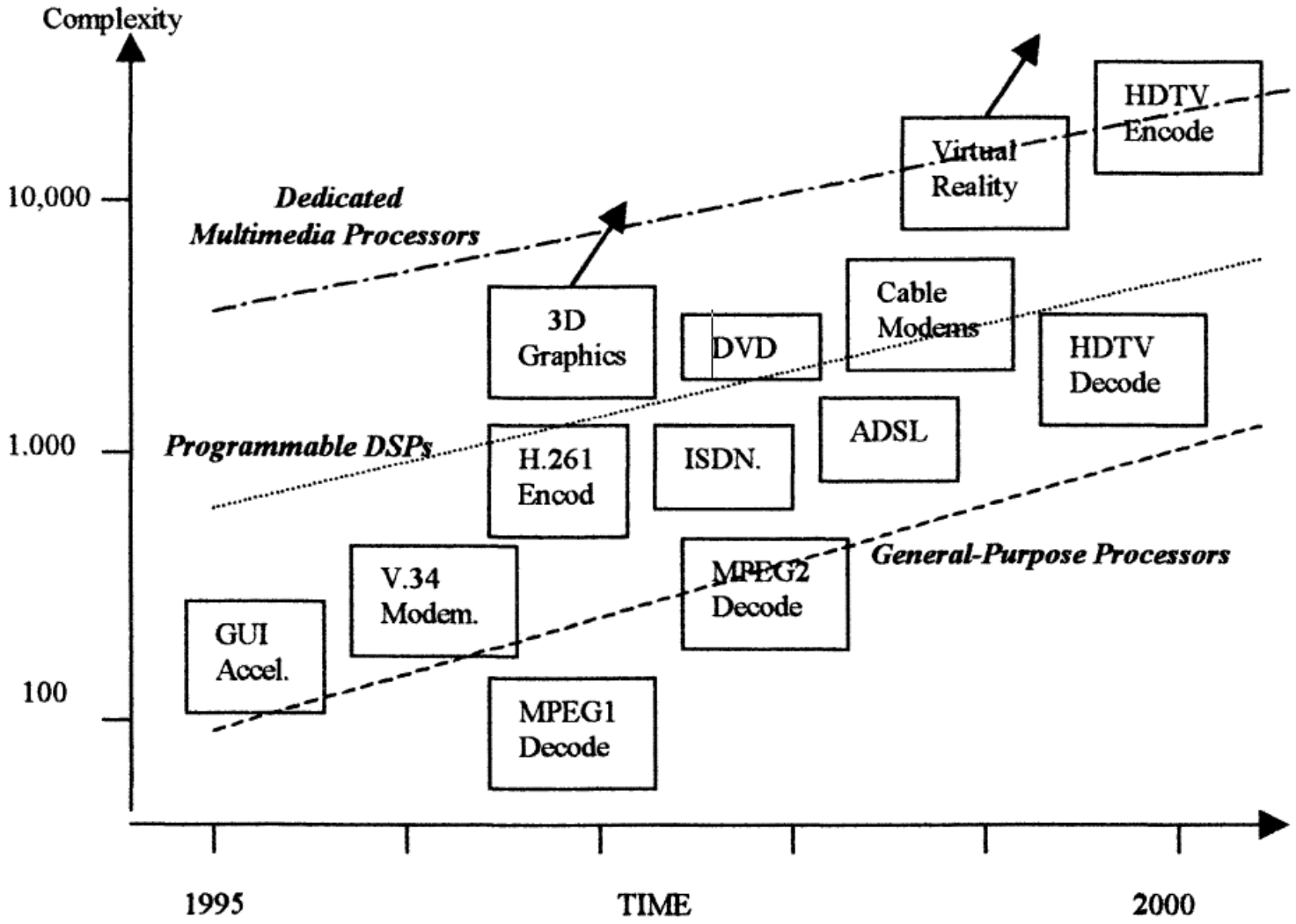
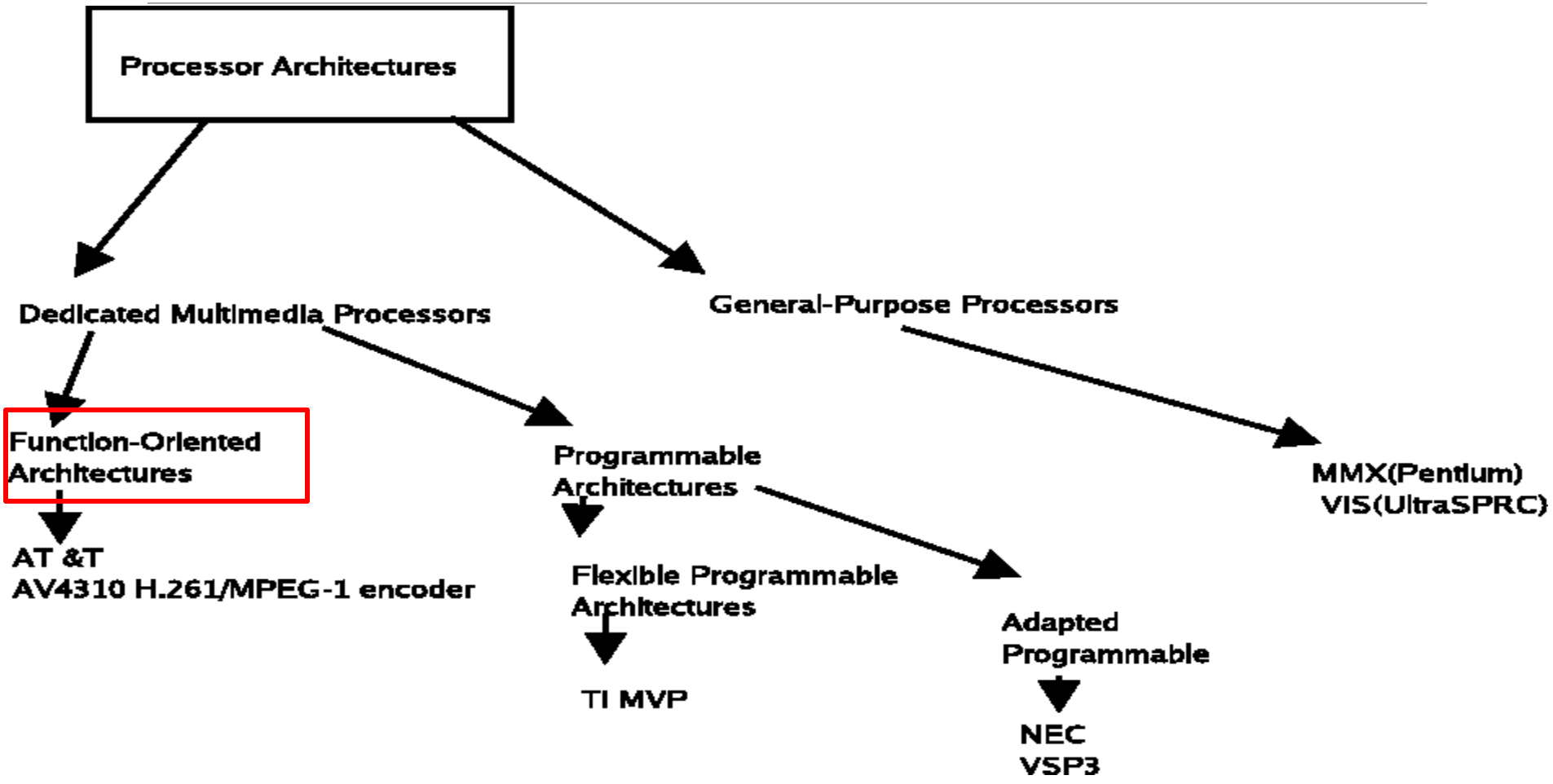


Figure 3. MOPS requirements for various multimedia funcions and current trends in computing power.

Dedicated multimedia processors



Function Specific Dedicated Multimedia Architectures

Function Specific Dedicated Multimedia Architectures

- Provide limited and specific encoding or decoding standard
- However, their efficiency and speed are typically better compared to programmable architectures.
- The silicon area optimization achieved by function specific architectures allows lower production cost.
- In the first generation of function specific video processors, each of these functions was implemented in one chip, and a chipset was necessary to create the system for encoding or decoding.
- However, the next generations of function specific architectures integrate at these functions in a single VLSI chip.

Function Specific Dedicated Multimedia Architectures

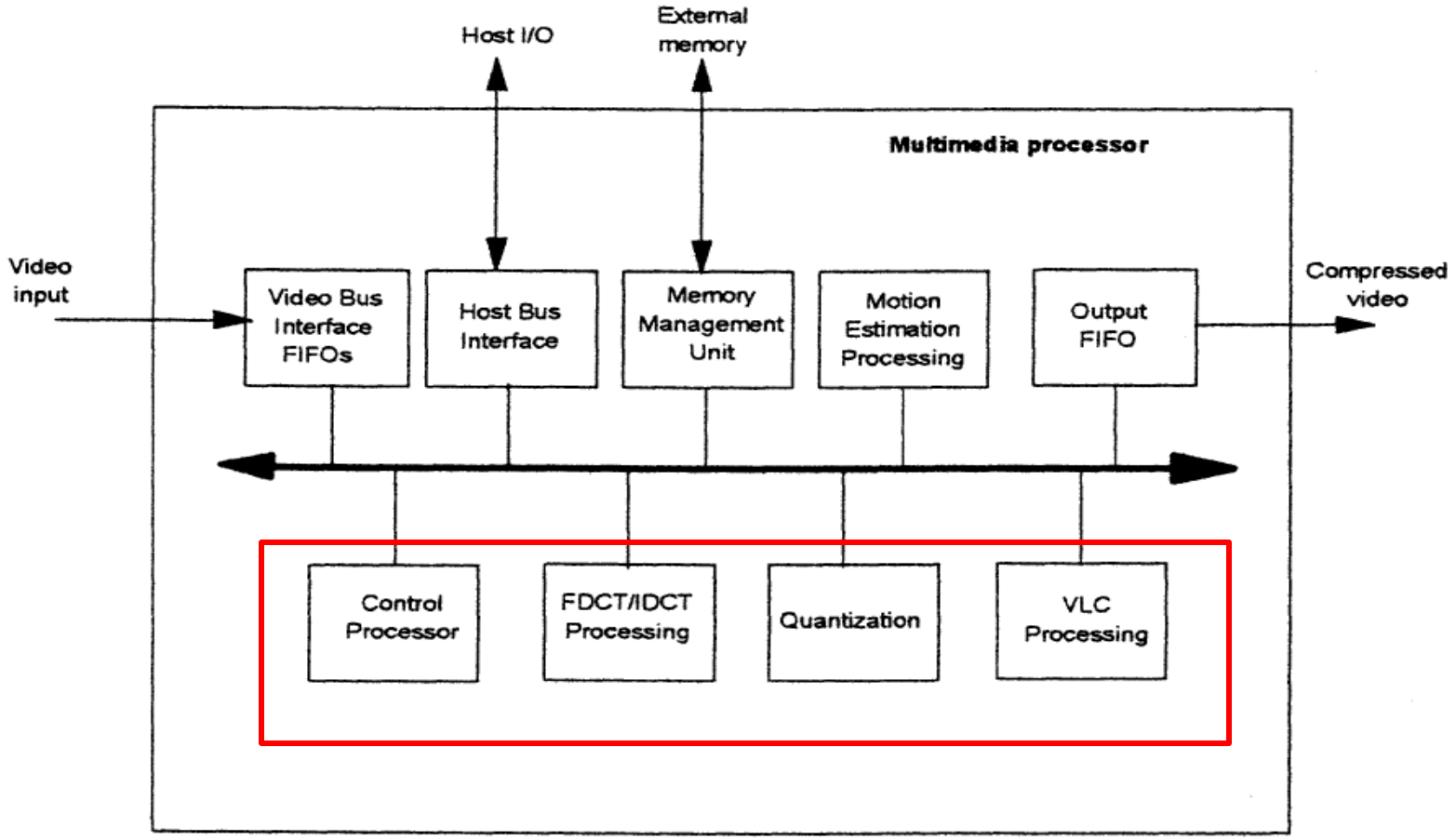
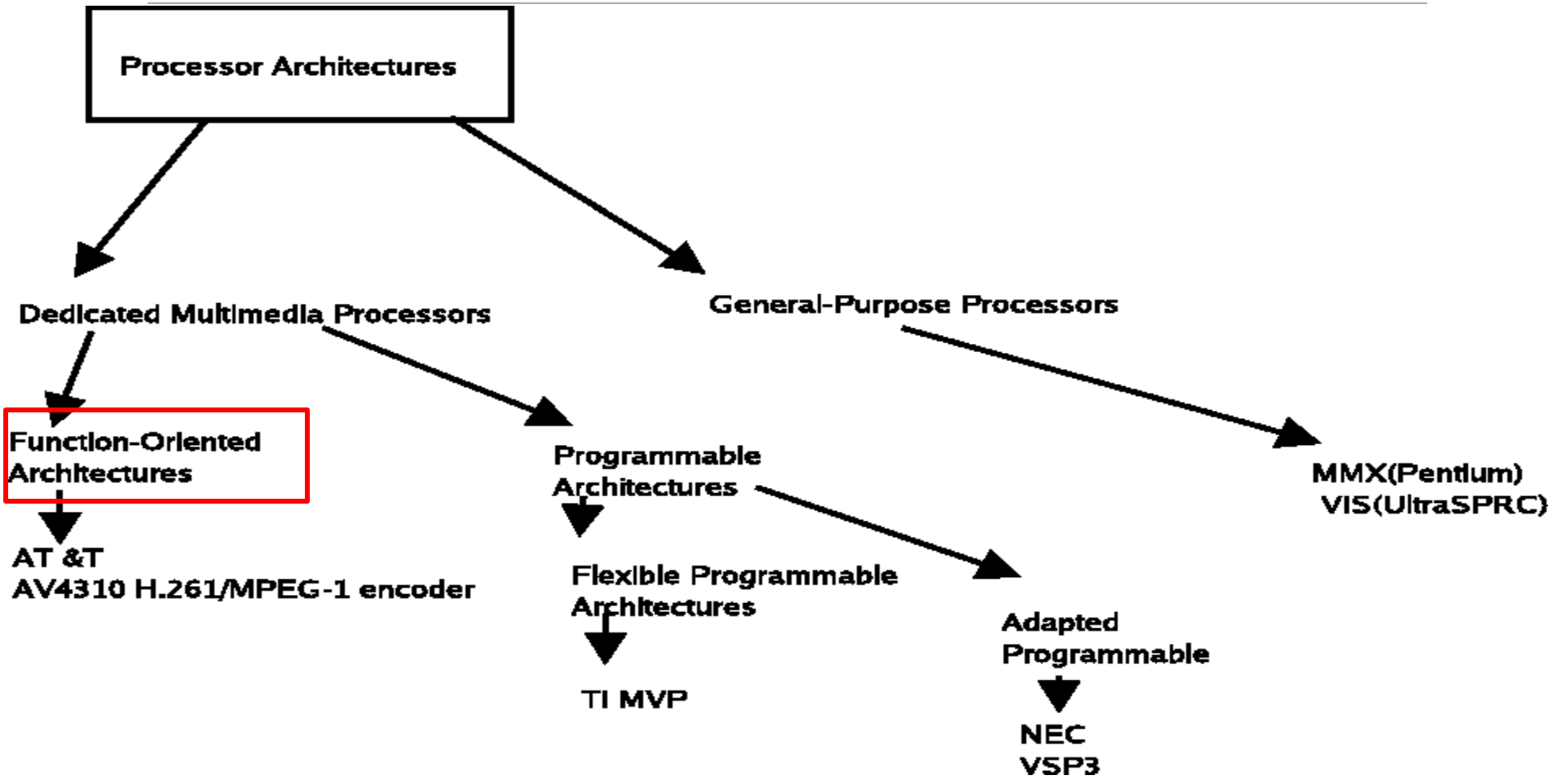


Figure 4. Block diagram of a typical function specific architecture for a video encoder. The dedicated processors (or functional units) are used for various operations, such as DCT, quantization, variable length coding (VLC), motion estimation, etc.

Programmable Dedicated Multimedia Architectures

Dedicated multimedia processors

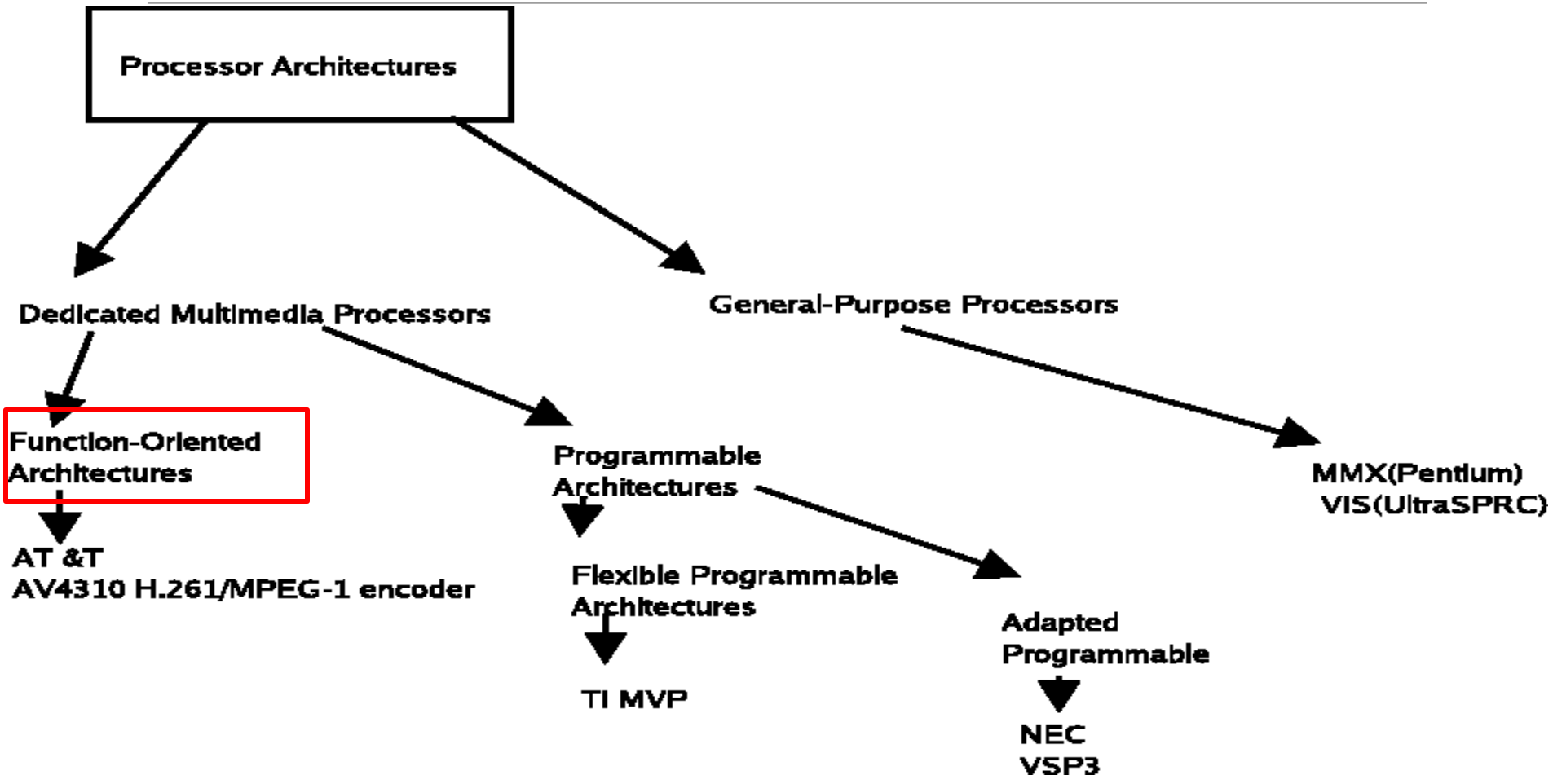


Programmable Dedicated Multimedia Architectures

- Programmable architectures enable the processing of different tasks under **software control**.
- The main advantage of programmable architectures is the increased flexibility.
- Changes of architectural requirements, such as changes of algorithms or an extension of the application domain, can be handled by software changes.
- Require a higher cost for design and manufacturing, since additional hardware for program control is required.
- Require software development for the application. Video coding applications require real-time processing of the image data, and therefore parallelization strategies have to be applied.

Flexible
Programmable
Dedicated
Multimedia
Architectures

Dedicated multimedia processors



Flexible Programmable Dedicated Multimedia Architectures

- **Flexible programmable architectures**, with moderate to high flexibility, are based on **the coprocessor concept as well as parallel data paths** and deeply pipelined designs.
- An example of a commercially available video processor, based on flexible programmable architecture, is TI's Multimedia Video Processor (MVP), TMS320C80.
- The MVP combines a RISC master processor and four DSP processors in a crossbar-based SIMD shared-memory architecture.
- The master processor can be used for control, floating-point operations, audio processing, or 3D graphics transformations.
- Each DSP performs all the typical operations of a general-purpose DSP and can also perform bit-field and multiplexing operations.



Flexible Programmable Dedicated Multimedia Architectures

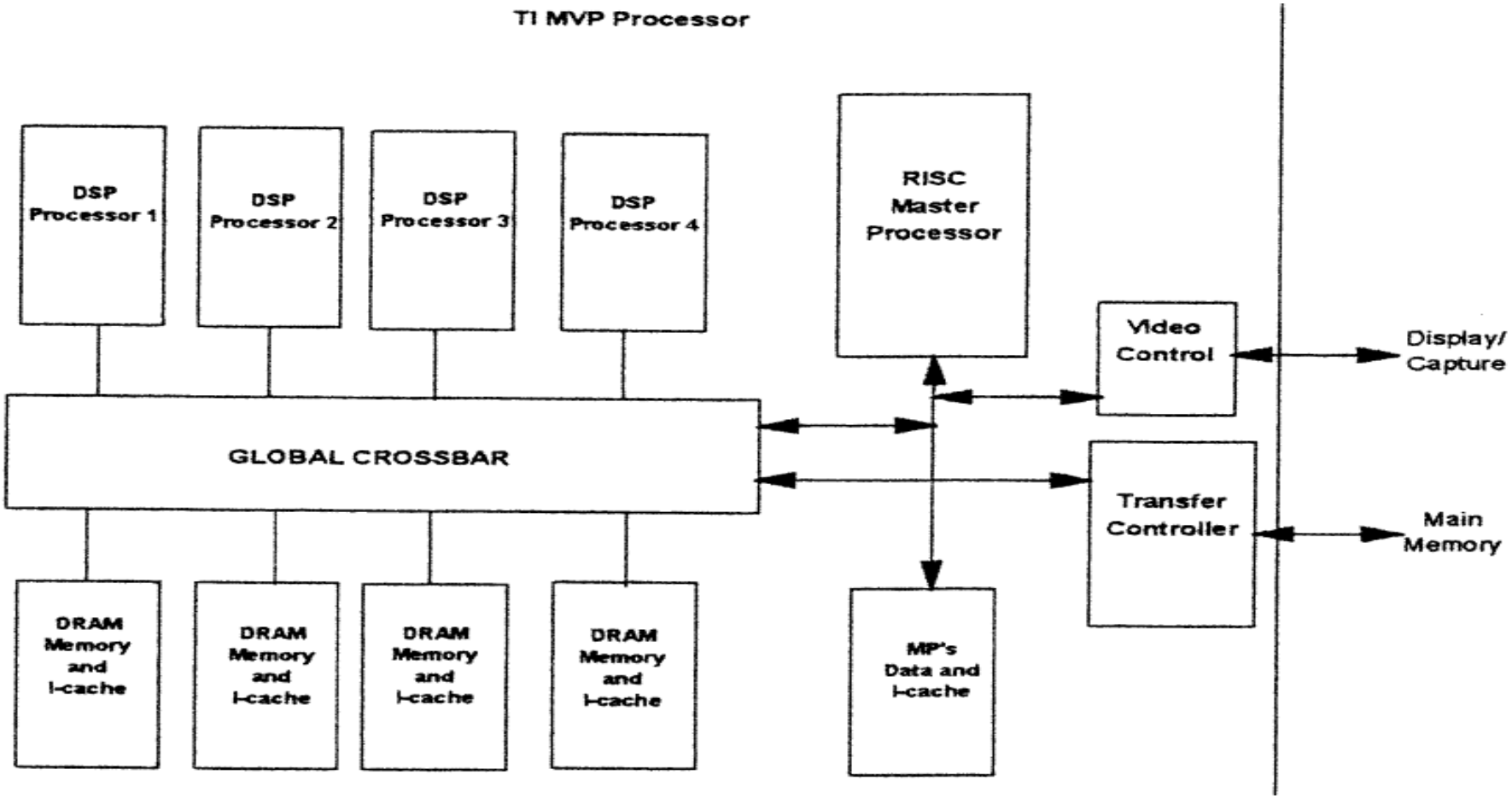


Figure *5.* Block diagram of the TI's Mulimedia Video Processor

Flexible Programmable Dedicated Multimedia Architectures

The MVP has been integrated into the Media station 5000, programmable multimedia system.

Its key function in this system is MPEG compression.

The data flow in the system during MPEG compression is shown in Figure 6.:

- Video data is captured into the video buffer at a resolution of 320x240.
- The MVP reads the data from the video buffer and stores it in the main (DRAM) memory.
- The MVP performs all MPEG compression functions on the data stored in the main memory.

Flexible Programmable Dedicated Multimedia Architectures

- Similar operations are performed on the digitized audio samples.
- Once the MVP completes the compression of a video or audio frame, the compressed bit stream is sent to the host computer.
- This is where the audio and video streams are multiplexed together, synchronized, and stored on a disk or transferred to a network
- Performance results show that the MediaStation 5000 system can achieve a real-time compression (30 fps) of MPEG-1 video sequences with resolutions of 320x240 (SIF format).
- Multiple MVPs are needed for real-time MPEG-2 encoding.

Flexible Programmable Dedicated Multimedia Architectures

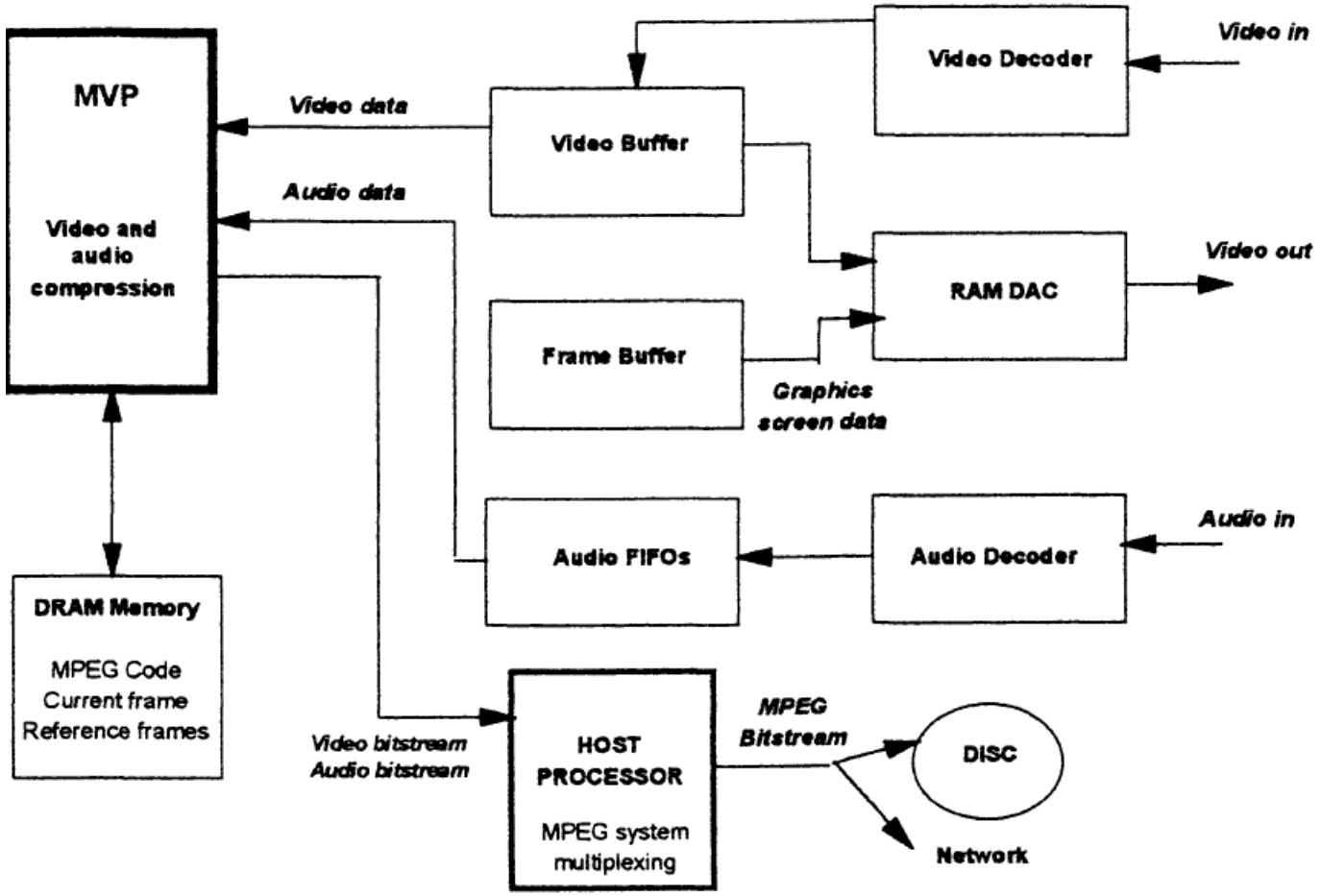
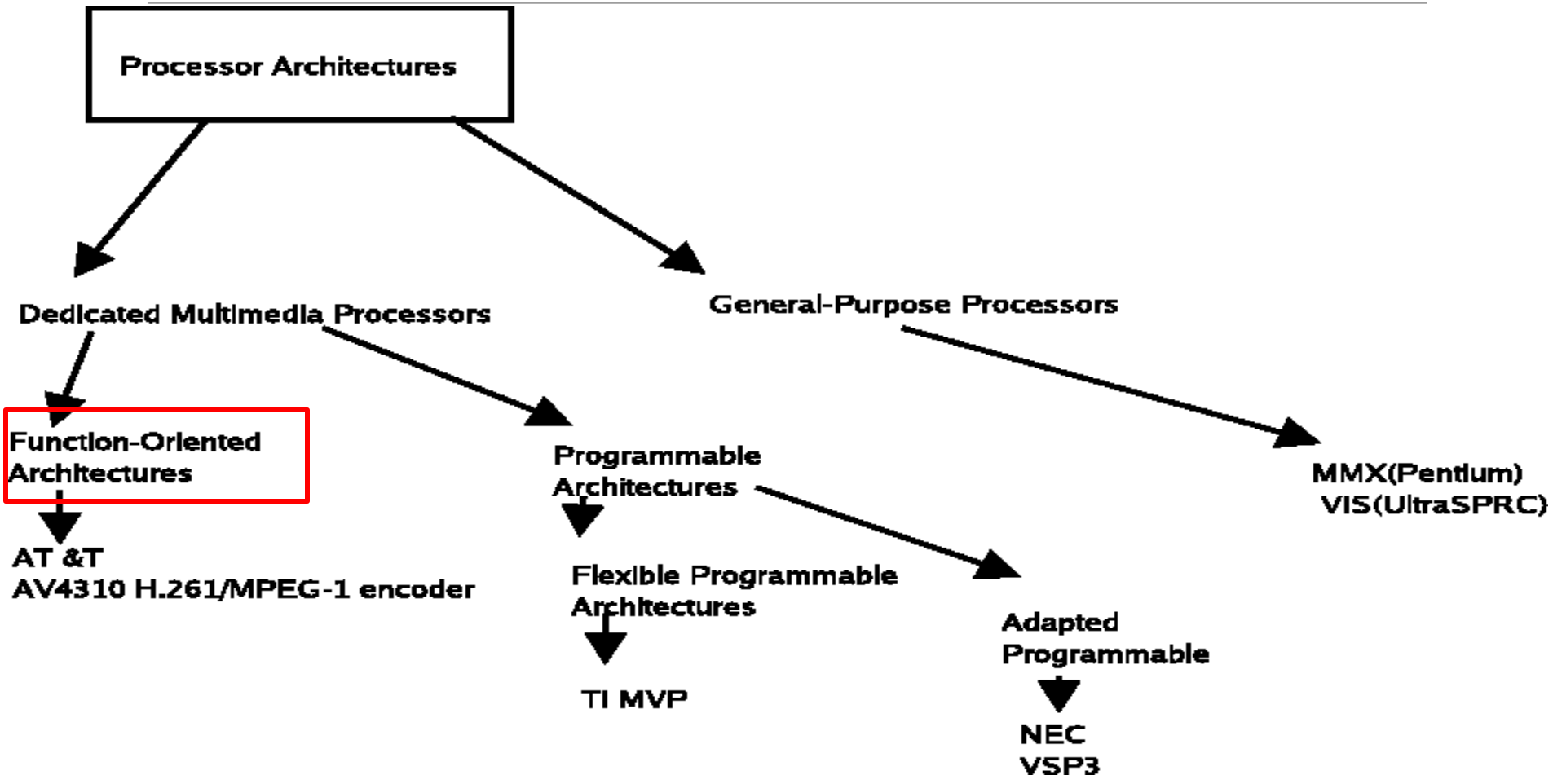


Figure 6. Data flow in the MVP during the MPEG compressicn

Adapted
Programmable
Dedicated
Multimedia
Architectures

Dedicated multimedia processors

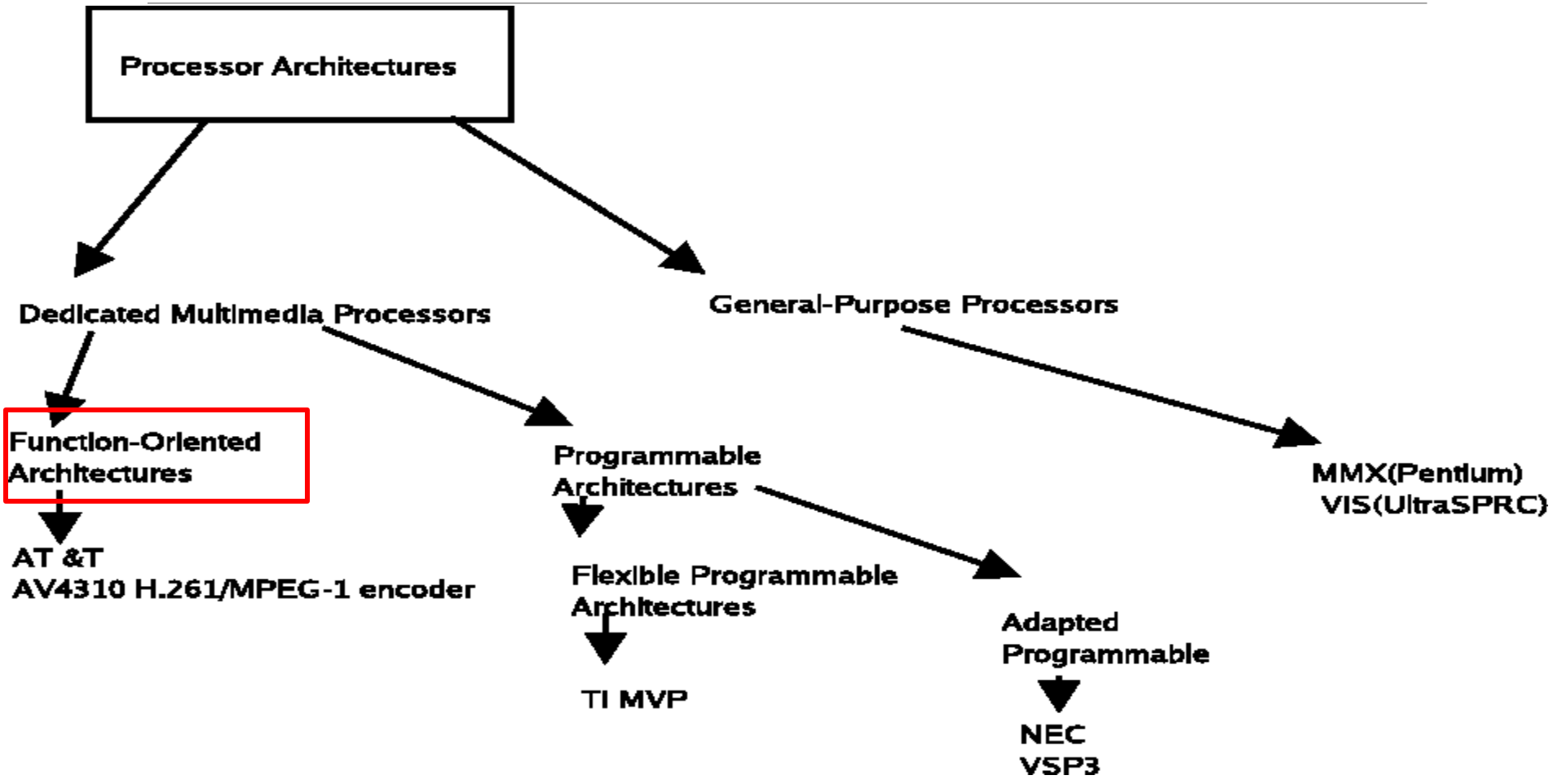


Adapted Programmable Dedicated Multimedia Architectures

- provide increased efficiency by adapting the architecture to the specific requirements of video coding applications.
- These architectures provide dedicated modules for several tasks of the video codec algorithm, such as DCT module, or variable length coding.
- Examples of a commercially available multimedia processor based on adapted programmable architecture are VideoRISC processors (VRP and VRP2) from Ccube Microsystems.

General-Purpose Multimedia Architectures

Dedicated multimedia processors



General-purpose (GP) processors

- Provide support for multimedia by including multimedia instructions into the instruction set.

- Instead performing specific multimedia functions (such as compression and 2D/3D graphics), **GP processors provide instructions specifically created to support generic operations in video processing.**

- For example, these instructions include support for:

- 8-bit data types (pixels)
- efficient data addressing
- I/O instructions, and even instructions to support motion estimation.

General-purpose (GP) processors

- The latest processors, such as MMX (Intel), VIS (Sun) and MAX-2 (HP), incorporate some types of SIMD architectures, which perform the same operation on multiple data elements in parallel.



New Architectural Trends

The latest dedicated processors use SIMD and VLIW architectural schemes and their variations to achieve very high parallelism.

The next generation of programmable multimedia processors incorporates increased parallelism by combining simd, vliw, and some other hybrid architectural schemes.

New Architectural Trends cont., SIMD architecture

- The SIMD parallel computer organization, applied in multimedia processors, typically uses a single control unit (or master processor), a number of processing elements (PEs) and shared memory among the PEs, as shown in Figure 8 .
- An interconnection network. such as crossbar switch, is used to interconnect the control processor, all PEs, and shared memory.
- The control processor evaluates every instruction. lf it is a scalar or program control operation, the scalar processor will directly execute it.

New Architectural Trends cont., SIMD architecture

- If the instruction is a vector operation, it will be broadcast to all the PEs for parallel execution.
- Partitioned data sets are distributed to the shared memory modules before starting the program execution.
- Then, the same instruction is executed by all the PEs in the same cycle, but on different data elements but on different data elements.

New Architectural Trends cont., SIMD architecture

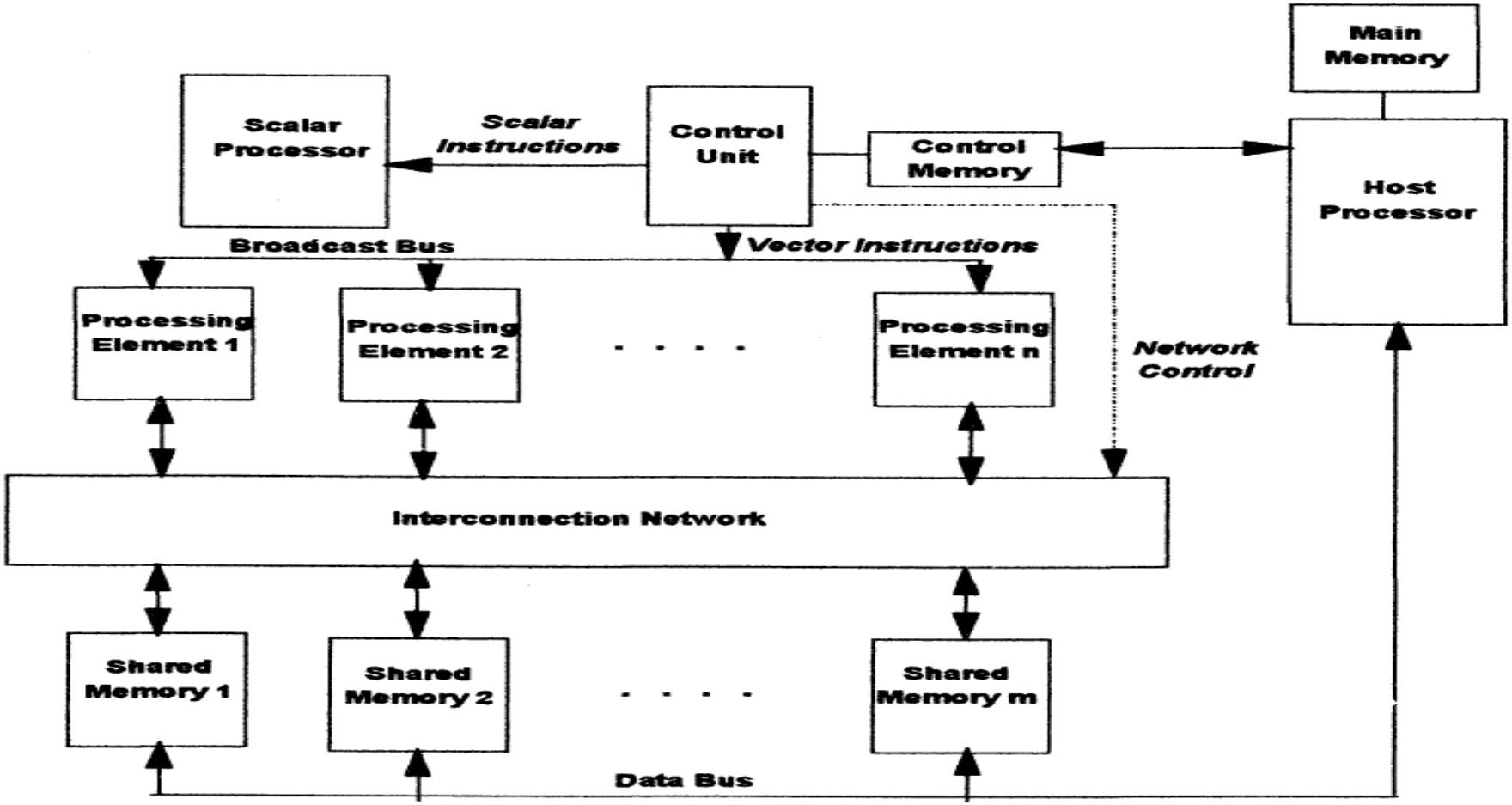


Figure 8. A general SIMD architectural model applied in multimedia processors

New Architectural Trends cont., VLIW architecture

- The VLIW architectural model is used in the latest dedicated multimedia processors.
- A typical VLIW architecture uses long instruction words with more than hundreds of bits in length.
- The idea behind VLIW concept is to reduce the number of cycles per instruction required for execution of highly complex and parallel algorithms by the use of multiple independent functional units that are directly controlled by long instruction words.
- All functional units share a common large register file. Different fields of the long instruction word contain opcodes to activate different functional units.
- Programs written for conventional 32-bit instruction word computers must be compacted to fit the VLIW instructions. This code compaction is typically done by a special compiler, which can predict branch outcomes, called trace scheduling.

New Architectural Trends cont., VLIW architecture

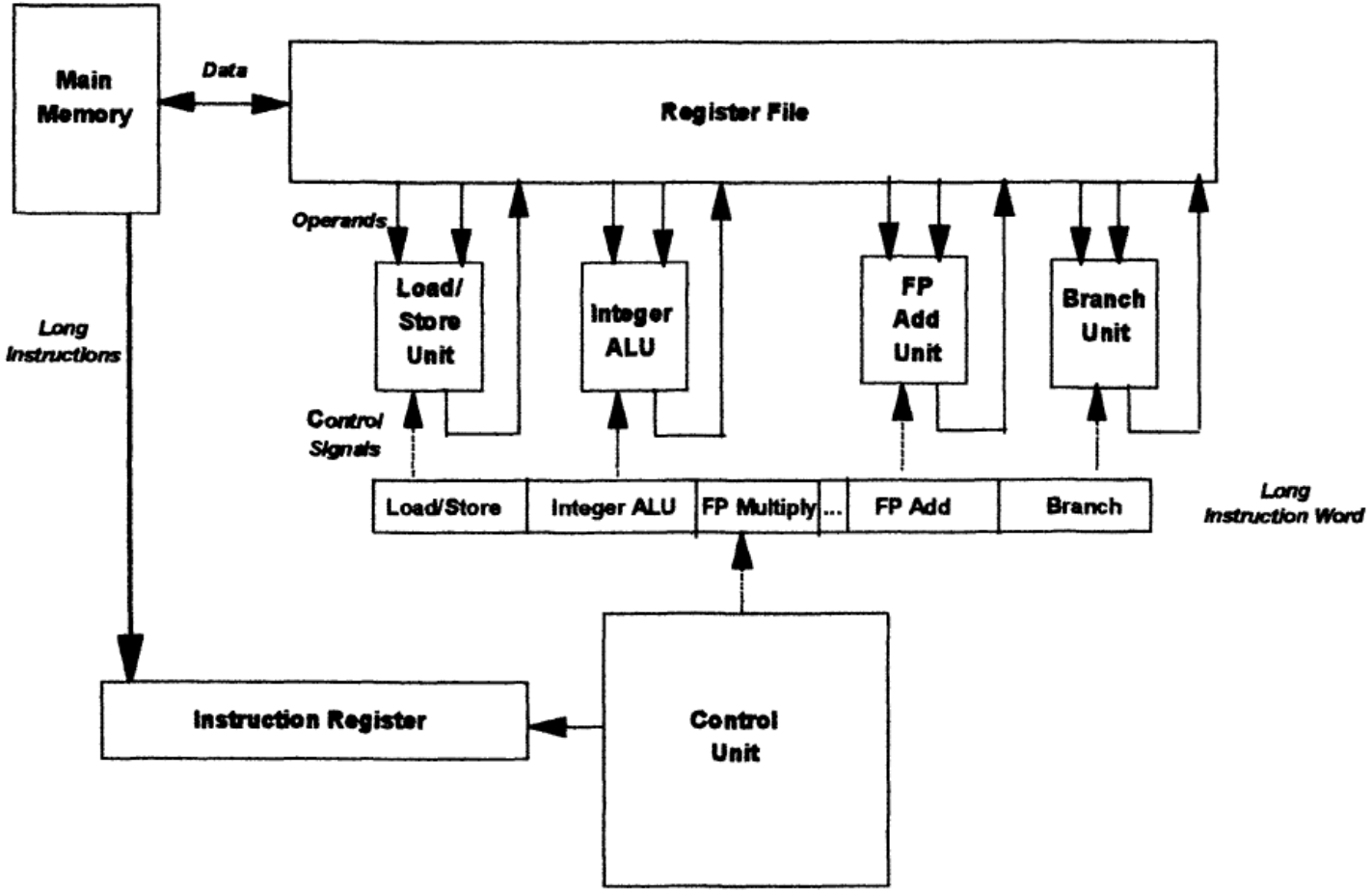


Figure 9. The very-long-instruction-word (VLIW) architectual model

Questions

Differentiate between the following according to (function, applications, and advantages).

1. Dedicated and general-purpose processors
2. Function-oriented architecture and programmable architecture.
3. Flexible programmable architecture and adapted programmable architecture.

Thank You